DELAGUA IMPROVED COOKSTOVE GROUPED PROJECT

DOCUMENT PREPARED BY DELAGUA

Project title	DelAgua Improved Cookstove Grouped Project
Project ID	3699
Crediting period	04-November-2022 to 03-November-2029
Original date of issue	04-November-2024
Most recent date of issue	21-August-2025
Version	11.0
VCS Standard Version	4.7
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1 PROJECT DETAILS

1.1 Summary Description of the Project

Summary description of the technologies/measures to be implemented by the project:

The clean cooking grouped project 3699 is a voluntary Grouped Project (GP) aimed at distributing fuel efficient improved cookstoves (ICS) to households in Rwanda. The Project Participant (PP) of the GP for all the communication, management, and ownership rights of the Greenhouse Gas (GHG) Emission Reduction Units (ERU) is DelAgua Trans Africa Stoves. Pte. Ltd. (DATAS) and the ICS under this GP will be distributed by DATAS. The distribution of ICS is not mandated by Rwandan law and the Grouped Project is a voluntary initiative being run by the PP. The program is not being implemented as a result of any government legislation or intervention. Throughout this Project Description, this grouped project may be interchangeably referred as “program". The PP plan to distribute 500,000[1] improved cookstove under the grouped project during first renewable crediting period. Each household will receive only one unit of ICS.

The location of the project:

Under the GP, PP will distribute the project technologies to individual households and communities within the host country Rwanda. The details of the GP and PAI locations are provided in Section 1.12 of this document.

An explanation of how the project is expected to generate GHG emission reductions or removals:

The VCS GP will reduce the firewood usage at household level, thereby reducing the GHG emissions compared to the baseline. The fuelwood savings will act as the main source of VCU generation for this project. The implementation (distribution of ICS units) and parameters set for ER calculations within the GP will be monitored on a periodic basis, based on the requirements provided within the applied methodology.

A brief description of the scenario existing prior to the implementation of the project:

The scenario existing prior to the implementation of the project activity would be usage of 3 stone fire and traditional stoves, which have a very low thermal efficiency, and therefore consume a higher amount of firewood as fuel for cooking.

An estimate of annual average and total GHG emission reductions and removals:

The average annual and total GHG emission reduction expected from the grouped project is expected to be 1,250,568 and 8,753,978 tCO_2e, respectively, over the first 7-year crediting period.

[1] This plan is tentative and subject to change based on actual implementation, influenced by market demand and user preferences, deemed outside the control of the PP, and shall not be considered a deviation from the project's description. The number of ICS units outlined above serves as an approximate figure for ex-ante calculations and is non-binding.

1.2 Audit History

Audit Type	Period	Program	VVB Name	Number of years
Validation	04-November-2022 - 03-November-2029	VCS	Earthood Services Private Limited	7 years (1st CP)
Verification	04-November-2022 - 31-December-2022	VCS	Earthood Services Private Limited	0.158 year (58 days)
Verification	01-January-2023 – 31-December-2024	VCS	*SustainCERT SA*	Two Years

1.3 Sectoral Scope and Project Type

Sectoral scope[2]	03 -Energy demand
Project activity type	Type II - Energy efficiency improvement projects

1.4 Project Eligibility

1.4.1 General eligibility

The Grouped Project Activity falls within the scope of the VCS Program. This is elaborated under the tables below.

Table 1: VCS Program scope

Scope	Applicability
The seven Kyoto Protocol greenhouse gases.	The GP reduces GHG emissions associated with the combustion of cooking fuels. Gases included are CO_2, CH_4, N_2O.
Ozone-depleting substances (ODS).	The GP does not involve any ODS.
Project activities supported by a methodology approved under the VCS Program through the methodology development and review process.	The ICS distributed (& to be distributed) in the GP applies the methodology "VM0050 version 1.0." which is a VCS approved methodology.
Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval.	Not applicable.
Jurisdictional REDD+ programs and nested REDD+ projects as set out in the Jurisdictional and Nested REDD+ (JNR) Requirements.	Not applicable.

[2] Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes

The scope of the VCS Program excludes projects that can reasonably be assumed to have generated GHG emissions primarily for the purpose of their subsequent reduction, removal, or destruction.	NA, the grouped project reduces existing emissions from traditional biomass stoves, which would persist without intervention. Improved cookstoves lower fuel use, cut GHG emission, improve health. Unlike excluded projects, it addresses an existing issue, ensuring genuine and verifiable emission reductions.

Table 2: Excluded project activities under the VCS Program

Excluded Activity	Applicability
Grid-connected electricity generation activities using hydroelectric power plants.	N/A The GP reduces GHG emissions associated with the combustion of cooking fuels.
Grid-connected electricity generation activities using wind, geothermal, or solar photovoltaic (PV) power plants.	N/A. The GP activities are energy efficiency measures and do not generate electricity from wind, geothermal, or solar power plants/units.
Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration.	N/A The GP activities are energy efficiency measures and do not utilize recovered waste heat for any purposes.
Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass.	N/A. The GP activities are energy efficiency measures and do not generate electricity using biomass. The GP includes energy efficiency improvements in thermal applications, i.e. cookstoves.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel.	N/A. The GP activities do not include the generation of electricity and/or thermal energy using fossil fuels, including activities that involve switching from a higher carbon content fuel to a lower carbon content fuel.
Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs).	N/A The GP activities do not include the replacement of electric lighting.

Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers.	N/A. The GP activities do not include the installation and/or replacement of electricity transmission lines and/or energy efficient transformers.
Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	N/A. The GP reduces GHG emissions associated with the combustion of cooking fuels

Requirements related to the pipeline listing deadline, the opening meeting with the validation/verification body, and the validation deadline:

Reference to VCS Standard version 4.7	Requirement	Justification
3.8.1	Non-AFOLU projects shall complete validation within two years of the project start date. Additional time is granted for non-AFOLU projects to complete validation where they are applying a new VCS methodology. Specifically, projects using a new VCS methodology and completing validation within two years of the approval of the methodology by Verra may complete validation within four years of the project start date.	The project is already registered under VCS. This version of the VCS PD corresponds to the application of VM0050 v1.0.
4.1.5	The validation/verification body shall ensure that the project is listed on the project pipeline with a status of under validation before the opening meeting with the project proponent, such opening meeting representing the beginning of the validation process.	The project is already registered under VCS. This version of the VCS PD corresponds to the application of VM0050 v1.0.

Eligibility of applied methodology under the VCS Program:

The project applies VM0050 (Energy Efficiency and Fuel-Switch Measures in Cookstoves, Version 1.0), an approved methodology under the VCS Program. As VM0050 does not impose any scale or capacity limits, the project is not subject to restrictions on its size or the number of cookstoves distributed. Additionally, the project is implemented as a Grouped Project with clearly defined geographic and technical criteria, ensuring it is not a fragmented part of a larger activity.

Compliance with the general requirements is described below:

Reference to VCS Standard version 4.7	Requirement	Justification
3.1.1	Projects shall meet all applicable rules and requirements set out under the VCS Program, including this document.	The grouped project meets all applicable rules and requirements as set out under the VCS Program and is guided by the ISO 14064-2:2006

	Projects shall be guided by the principles set out in Section 2.2.1.	
3.1.2	Projects shall apply methodologies eligible under the VCS Program. Methodologies shall be applied in full, including the full application of any tools or modules referred to by a methodology, noting the exception set out in Section 3.14.1.	The grouped project applies VCS approved methodology VM0050, version 1.0 along with tools or modules as applicable, ensuring full compliance.
3.1.3	Projects shall apply the latest version of the applicable methodology in all cases unless a grace period applies to the project as set out in 3.22. Projects shall update to the latest version of the methodology when reassessing the baseline or renewing a crediting period.	The grouped project applies the latest version 1.0 of VCS approved methodology VM0050. Refer to section 3 of the PD below for more details.
3.1.4	Projects and the implementation of project activities shall not lead to the violation of any applicable law, regardless of whether or not the law is enforced.	The grouped project and its instances shall remain in compliance with the applicable local laws. This has also been made an eligibility criterion for inclusion of project activity instances into the grouped project.
3.1.5	Where projects apply methodologies that permit the project proponent its own choice of model, such model shall meet with the requirements set out in the VCS Program document VCS Methodology Requirements and it shall be demonstrated at validation that the model is appropriate to the project circumstances (i.e., use of the model will lead to an appropriate quantification of GHG emission reductions or removals).	Not applicable. The PP has applied the model presented in the applied methodology VM0050 version 1.0
3.1.6	Where projects apply methodologies that permit the project proponent to choose a third-party default factor or standard to ascertain GHG emission data and any supporting data for establishing baseline scenarios and demonstrating additionality, such default factor or standard shall meet with the requirements set out in the VCS Program document VCS Methodology Requirements.	Not applicable. The applied methodology, VM0050 version 1.0, does not allow for the selection of alternative models. The project proponent has followed the prescribed model outlined in the methodology.
3.1.7	Where the rules and requirements under an approved GHG program conflict with the rules and requirements of the VCS	The project activity applies VCS methodology and so this is not applicable.

	Program, the rules and requirements of the VCS Program shall take precedence.	
3.1.8	Where projects apply methodologies from approved GHG programs, they shall conform with any specified capacity limits and any other relevant requirements set out with respect to the application of the methodology and/or tools referenced by the methodology under those programs.	The project activity applies VCS methodology i.e., VM0050 version 1.0 that does not specify any capacity limit.
3.1.9	Where Verra issues new VCS Program rules, the effective dates of these requirements are set out in Appendix 3 of VCS standard ver 4.7 and Effective Dates or equivalent for other program documents and are listed in a companion Summary of Effective Dates document which corresponds with each update.	The grouped project will adhere to the new VCS Program rules, and the effective dates of these requirements set out in Appendix 3 of VCS standard ver 4.7.

1.4.2 AFOLU project eligibility

Not Applicable, as this is not a AFOLU project.

1.4.3 Transfer project eligibility

Not Applicable, as this project is not seeking any transfer from other GHG reduction programs such as CDM.

1.5 Project Design

☐ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☒ Grouped project

1.5.1 Grouped project design

The project has been designed to include distribution of ICS units at multiple locations (districts, sectors, cells, and umudugudus) within the project boundary, under multiple project activity instances within a grouped project.

Eligibility Criteria

VCS 3699, being a Grouped Project Activity, will be implemented while fulfilling the eligibility criteria provided in the Table below.

Eligibility criteria as per section 3.6.10 to 3.6.15 of the VCS Standard v4.7

Criterion	How the new project activity instances comply	Evidence
Grouped projects shall specify one or more clearly defined geographic areas within which project activity instances may be developed. Such geographic areas shall be specified using geodetic polygons	Each PAI will have clearly defined geographic areas and such area will be specified using geodetic polygons.	PAI 1 has been implemented in Rwanda. PP has submitted geodetic polygons to VVB and added details in section 1.12 of the PD MR.
Determination of baseline scenario and demonstration of additionality are based upon the initial project activity instances. The initial project activity instances are those that are included in the project description at validation and shall include all project activity instances currently implemented on the issue date of the project description.	Baseline scenario and demonstration of additionality shall be based upon the initial project activity instances	Baseline scenario and additionality has been demonstrated for PAI1 which is included at the time of validation in section 3.4 and 3.5 of the PD MR.
Where a grouped project includes multiple project activities, the project description shall designate which project activities may occur in each geographic area.	The grouped project includes only one project activity.	The PAI-1 includes distribution of improved cookstoves to households that are dependent on traditional cookstoves using woody biomass for cooking.
The baseline scenario for a project activity shall be determined for each designated geographic area, in accordance with the methodology applied to the project.	Baseline scenario for a project activity shall be determined for each designated geographic area, in accordance with the methodology applied to the project	Baseline scenario has been determined for PAI-1 for Rwanda. Details have been added in section 3.4
The additionality of the initial project activity instances shall be demonstrated for each designated geographic area, in accordance with the	Additionality for a project activity shall be determined for each designated geographic area, in accordance with the	Additionality has been determined as per the designated methodology. Details have been added in section 3.5.

methodology applied to the project	methodology applied to the project	
Where factors relevant to the determination of the baseline scenario or demonstration of additionality require assessment across a given area, the area shall be, at a minimum, the grouped project geographic area. Examples of such factors include, inter alia, common practice; laws, statutes, regulatory frameworks, or policies relevant to demonstration of regulatory surplus; determination of regional grid emission factors; and historical deforestation and degradation rates	All the factors relevant to the determination of baseline or demonstration of additionality shall be done for the geographical boundary of the grouped project	Assessment of baseline and demonstration of additionality for PAIs has been done for Rwanda.

Eligibility criteria as per section 3.6.16 and 3.6.17 of the VCS Standard v4.7:

S No	Criterion	How the new project activity instances comply	Evidence
1	Project Activity Instances (PAIs) must meet the applicability conditions set out in the applied methodology.	All the PAIs must meet the applicability conditions set out in the applied methodology as defined in section 3.2 below.	The Initial PAI-1 meets all applicability conditions for methodology (VM0050 version 1.0) as defined in section 3.2 below.
2	Use the technologies or measures specified in the project description.	Only ICS that conform with the GP description are to be distributed in the project. The ICS will be chosen to deliver a level of service at least equivalent to the baseline appliance.	Manufacturer's specifications. The ICS deployed under PAI-1 is Kuniokoa Generation 2 with Pot Skirt, which is as per technology description of the grouped project.
3	Apply the technologies or measures in the same manner as specified in the project description.	The ICS distributed in the PAIs will adhere to the GP description (i.e. burning of biomass for cooking meals) and replace the baseline device which also was	Manufacturer's specifications. The ICS distributed under PAI-1 reduces fuel wood consumption due to better thermal efficiency

		utilising the biomass burned in the baseline scenario.	and replaces the baseline traditional stoves, the same can be checked with distribution record and sample survey results.
4	Are subject to the baseline scenario determined in the project description for the specified project activity and geographic area.	All new PAIs will be implemented only in regions within the geographic borders of Rwanda subject to the same baseline scenario determined in Section 3.4 in this Project Description. The geographical location of the project ICS can be traced through the information provided in the distribution database, which includes the location data of the ICS distributed.	ICS Distribution database For PAI-1, records distribution (end user registration database) with clear delineation of project boundary i.e. host country of Rwanda is provided to the VVB for their reference.
5	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	All new PAIs will use the activity method for demonstration of additionality. Step 1: Regulatory Surplus There is no government mandated programme or policy in host country of this project ensuring the distribution of new project activity instances. Step 2: Positive List The inclusion of new project activity instances must meet both of the following conditions to qualify for the positive list: 1) The project activity introduces:	Step1: Regulatory Surplus There is no mandatory legal requirement for installation of cookstoves in households/ institutions of Rwanda. Step 2: Positive List All improved cookstoves being distributed within PAI1 are non-renewable, efficient biomass-fired cookstoves that replace inefficient biomass-fired cookstoves in Rwandan households. These ICS are distributed at zero cost to the end user. Thus, the project complies with both Step 1 and Step 2, and therefore, is

		a) Efficient biomass-fired cookstoves that replace inefficient biomass-fired cookstoves; b) Efficient, solely renewable biomass-fired cookstoves that replace fossil fuel-fired cookstoves; or c) Electric cookstoves that replace inefficient biomass-fired or fossil fuel-fired cookstoves. 2) The project activity installs or distributes improved cookstoves at zero cost to the enduser and has no revenue source other than from the sale of verified carbon units (VCUs). All the new PAIs that pass the regulatory surplus test (Step 1) and qualify for the positive list (Step 2) are deemed additional and are not required to apply Step 3.	deemed additional. The above can be checked with end user agreement and declaration from PP on the same. The PP has provided end user agreement and declaration to VVB to support free distribution of ICS.
Eligibility conditions specific to inclusion of New PAIs			
6	The new PAI must occur within one of the designated geographic areas specified in the project description	The distribution of the ICS will be done in the jurisdictions of Rwanda. Each PAI distributes the ICS units to HHs in umudugudus (Kinyarwanda term used for villages) and cells which are distinct from each other.	1. Sample list of the beneficiaries that are to be provided the project ICS will be shared. 2. Project ICS distribution data, recorded during the HH visit. For PAI-1, records distribution (end user

		No two PAI or VCS Grouped projects in the host country Rwanda can have ICS units distributed to same HH. This is ensured by accessing the beneficiary list from the cell office, which is then verified on-site by the DelAgua team for any corrections in the number of beneficiaries. Post that, the final list of beneficiaries is reviewed by the umudugudu (village) chief. Distribution related information of ICS units specific to each PAI and other DelAgua grouped project activities is then recorded in separate databases using electronic data recording software, form.com. The unique identifiers of the distributed ICS unit reflect that it is distributed to a unique HH in the host country Rwanda.	registration database) with clear delineation of project boundary i.e. host country of Rwanda is provided to the VVB for their reference.
7	The new PAI must comply with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial compliance with multiple sets of eligibility criteria is insufficient.	The eligibility criteria set must be complied by the PAIs at all points in time throughout the crediting period. If at any point, it is found that the PAI is not complying with the set of eligibility criteria, then it may be excluded from the crediting for the duration of the monitoring period, until it is able to demonstrate	The MR will act as the documentary compliance summary. However, for each specific eligibility criteria (as laid down above), the supporting evidence will be provided by the PP. The PAI-1 is part of VCS grouped project validation/verification,

		the compliance of the eligibility criteria.	hence no further assessment.
8	The new PAI must be included in the monitoring report with sufficient technical, financial, geographic and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.	The new PAI will include all the technical, financial, geographic and other relevant information related to the compliance of the applicable set of eligibility criteria & also enable sampling by the VVB.	The MR for the specific monitoring period will act as the supporting evidence for the stated condition/requirement. The PAI-1 is part of VCS grouped project validation/verification, hence no further assessment.
9	The new PAI must have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).	The PAI related evidence on the project ownership will be in the form of a distribution end user receipt poster. This will be signed by the end user, and a photograph of him/her holding the poster along with the ICS will be saved in the PAI database.	1. A photograph of the PAI distribution first end user holding the poster along with the ICS. 2. A scan of the poster, which will show the first distribution date. 3. The distribution database of the GP will provide the start date in the form of the first distribution made by the PAI. Waiver of carbon credits from the end users are submitted to the VVB for PAI-1.
10	The new PAI must have a start date that is the same as or later than the grouped project start date.	The new PAI will have a start date later than the GP start date.	This can be verified through the GP distribution database, wherein each new PAI can be verified for its start date. The PAI-1 is first instance and start date of grouped project activity is same as start date of PAI-1. The

			distribution record shared with VVB to support start date of PAI-1.
11	The new PAI must be eligible for crediting from the later of start date of the project activity instance or the start of the verification period in which they were added to the grouped project, through to the end of the project crediting period (only).	The PAI will only be eligible for crediting from the start date through to the end of the project crediting period (only).	1. The start date of the PAI can be checked through the distribution database. 2. The end date of the GP crediting shall be referenced from the VCS Project Registry webpage. 3. The overlap of monitoring periods may be checked through the previous monitoring reports (available publicly on VCS project registry webpage for those periods). The PAI-1, is first instance and start date of grouped project activity is same as start date pf PAI-1, hence no overlapping of crediting period.
12	The new PAI shall not leave one VCS project and subsequently enrolled in another VCS project.	It will be ensured that the PAI that is being enrolled under the VCS GP 3699 was not previously enrolled under any other VCS project, and will not be enrolled under other VCS project if it is removed from the VCS GP 3699.	This will be evidenced through a declaration signed by the PP for the PAIs that are enrolled under the VCS GP 3699. Since, for PAI-1 project proponent is the implementer, no further approval is required, however for the cases, where implementer other than PP, an approval on the same shall be given by PP in order to include the instance into this grouped project.

13	The new PAI shall adhere to clustering requirement that states the project proponent shall include in a singular project all project activity instances within ten kilometers of another instance of the same project activity and with the same project proponent	It will be ensured that PAI included within 10 kilometers of another instance of the same project activity by the same PP is in a singular project	PAI-1 involves distribution of improved cookstoves by same PP within 10 kilometers hence included in the same project. There are no other project instances by the same PP distributed within 10 kilometers for the same project activity.

1.6 Project Proponent

Organization name	DelAgua Trans Africa Stoves Pte. Ltd. (DATAS)
Contact person	Neil McDougall
Title	Mr.
Address	38 Beach Road, South Beach Tower, #29-11 Singapore 189767
Telephone	+352 661 939 304
Email	vcsprojects@delagua.org

1.7 Other Entities Involved in the Project

Organization name	DelAgua Health Rwanda (Implementation) Limited
Role in the project	Distribution, operational and monitoring services
Contact person	Rory McDougall
Title	Mr.
Address	#59 Kg 9 Ave, Gasabo, Kigali, Rwanda
Telephone	+250 788 386 128
Email	vcsprojects@delagua.org

1.8 Ownership

DelAgua Trans Africa Stoves Pte. Ltd. (DATAS) is the owner of the VCS grouped project activity.

During the distribution of the improved cookstoves, the participating households sign an end user agreement between customer and DATAS. The end user agreement has customer information, unique identification number, product details etc. The agreement follows the requirements of para 3.7.1 of the VCS Standard ver4.7 which states the evidence to establish project ownership should be "An enforceable and irrevocable agreement with the holder of the statutory, property or contractual right in the plant, equipment or process that generates GHG emission reductions and/or removals which vests project ownership in the project proponent."

The agreement confirms that the ownership rights of the VERRA project and the carbon assets generated from this project lie with the project proponent. The customer under the End User Agreement "releases all rights to the greenhouse gas reductions and carbon credits produced by the use of the clean energy product in the favour of DATAS and agree to not sell or transfer the GHG or carbon credits to any other third party or use these credits for any other purposes"

This will be evidenced through the below:

- Manufacturer agreements confirm that DATAS is the sole owner of all technologies purchased under this project. These agreements further confirm the manufacturer's understanding that the carbon credits associated with use of the technologies shall be within the jurisdictions of DATAS.
- During the distribution of the ICS, the participating household sign an End User Agreement to confirm that the ownership rights of the carbon assets generated from this project lie with the project proponent.

1.9 Project Start Date

Project start date	04-November-2022
Justification	Project start date is the date of distribution of 1st ICS under the project.

1.10 Project Crediting Period

Crediting period	☒ Seven years, twice renewable ☐ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	04-November-2022 to 03-November-2029

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☐ < 300,000 tCO2e/year (project)

☒ ≥ 300,000 tCO2e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)
04-November-2022 to 31-December-2022	65,693
01-January-2023 to 31-December-2023	1,045,259
01-January-2024 to 31-December-2024	1,015,981
01-January-2025 to 31-December-2025	1,335,051
01-January-2026 to 31-December-2026	1,428,583
01-January-2027 to 31-December-2027	1,412,267
01-January-2028 to 31-December-2028	1,358,443
01-January-2029 to 03-November-2029	1,092,701
Total estimated ERRs during the first or fixed crediting period	**8,753,978**
Total number of years	7
Average annual ERRs	**1,250,568**

1.12 Description of the Project Activity

Distribution plan:

The distribution of ICS will occur at centralized distribution points, where households from surrounding villages will be educated on the proper use and maintenance of the technologies. Shortly following the initial distribution, CHWs will visit each household that received a technology to deliver the training again, ensuring that all members of the household are aware of proper use and maintenance.

The ICS distribution is planned in close coordination with the local government offices, e.g. District Offices, Sector Offices, and so forth. There is a structured plan through which the distribution activity is scheduled and then implemented on ground.

Project activity description (including the technologies or measures employed) and how it/they will achieve net GHG emission reductions or removals.

The project activity included in this project description is the distribution and installation of high efficiency ICS systems in PAIs within Rwanda.

The ICS units will be based on a “rocket stove” design which reduce GHG emissions from biomass burning through improved combustion efficiency of wood fuel and decreased wood fuel consumption. The project stoves improve heat transfer to cooking pots through the insulated walls and design of the stove top when compared to traditional cooking methods over open- and three-stone fires. The project stoves directly decrease wood fuel demand and reduce the pressure from wood cutting within the project boundary. A 30-50% saving of firewood when compared to three-stone fire cooking has been reported in certain models. The time spent foraging for wood fuel is thereby also reduced, saving time for especially women to engage in other economic driven activities.

The grouped project includes dissemination of following woody biomass ICS[3], besides others. Specification of these example ICS is given below:

ICS Model	Kuniokoa	Ecoa wood
Name of the stove	Kuniokoa Generation 2	Ecoa wood Generation 3
Stove manufacturer	Burn	Burn
Stove Type	Natural Draft	Natural Draft
Stove Body	Stainless Steel	Stainless Steel
Stove height (cm)	32.1	28.5
Stove diameter (cm)	28.2	26.9
Thermal efficiency with pot skirt (%)	51.3%	52.0%
Thermal efficiency without pot skirt (%)	43.4%	45.5%
High Power Output (kW)	2.9	2.8
Stove lifetime (Years)	10	10
Reference Picture of the ICS	 	

List of main manufacturing/production technologies arrangements, systems and equipment involved.

Detailed application and information on the ICS technology that is going to be distributed under the new PAIs will be described in the upcoming MRs, elaborating the information under relevant section of the MR for the newly included PAIs. PP will source the best available technologies for

[3] The ICS listed are non-exhaustive. The ICS distributions are subject to potential customer demands and preferences. Other ICS models may be included in the Project subsequently, provided they meet the methodological requirement. This shall not be construed as a deviation or change from project description.

distribution under the GP. The project already distributed 335,314 till the end of year 2024. The ICS model selection at PAI level will be based on key attributes such as thermal efficiency, durability, cultural acceptability, and manufacturing quality and warranty. From time to time, requirement gathering will be done on a continuous basis, seeking comments from the end users (through monitoring surveys, feedback requests on call center, feedback via register available at different sites, and anonymous feedback through CHWs), and based on these user feedbacks, newer models will be introduced to the GP, and new PAIs will be designed to accommodate for their variability in terms of design, specifications, and geographies within the host country (wherever required).

It will be verified by the PP that the project technologies are, at a minimum, meeting the technical specifications and requirements set in the respective methodologies (under the eligibility criteria).

At present, DelAgua is sourcing the ICS from different manufacturers, deploying them at its central warehouse through imports, and then distributing them in a scheduled manner. Each distribution involves various steps, which are elaborated in the relevant section of this document. In this whole process, quality of the equipment being distributed will be checked on spot basis. Each lot of the distribution batch will undergo spot checks, as it is an integral part of Quality Assurance at DelAgua. The ICS units that will be found as faulty, will not be distributed to the end users. At present, only rocket stove models are planned for the upcoming distribution PAIs, as their distribution is found both beneficial to the locals (usage and maintenance) as well as a successful one from the implementation point.

The types and levels of service

Level of Service: PAIs will provide ICS based on the ‘rocket stove’ design, delivering a level of service at least equivalent to the baseline appliance (open fire used for household cooking).

Type of Service: The ICS technology will deliver a specified high-power thermal efficiency of at least 25% and evidenced by a manufacturer’s specification and WBT certificate. This is a significant improvement on the baseline open fire thermal efficiency default of 15%.

List of facilities, systems and equipment in operation under the existing scenario prior to the implementation of the project.

Under the baseline scenario, households use non-renewable woody biomass for their fuel needs for cooking. The baseline appliance being replaced by the project is an open fire (typically based on the three-stone concept).

1.13 Project Location

Host Country: Rwanda

Province: All Provinces of Rwanda

District/Sector/Cell/Village etc.: All Districts; Sectors; Cells; and Village in Rwanda

Physical/Geographical location:

For each PAI, the project boundary will be defined as being within the geographic borders of Rwanda. The project boundary for each Project Activity Instance is the physical, geographical location of the ICS distributed under it within the geographical boundaries of Rwanda.

Also, each individual distribution under the project technology will record the geographical location related data (e.g. geographical information in-line with the government data), which can be verified from the recorded data (distribution data – distribution records, and household visit survey forms).

The fuel used in both the project and baseline devices is collected or purchased locally from the nearby fields or nearby markets within the project boundary. The Geo coordinates for the grouped project is as follows:

2° 50' 23.8992'' S and 28° 51' 42.1452'' E

1° 2' 50.82'' N and 30° 53' 56.6592'' E

The map of Rwanda, for reference purpose, is as follows:

KML files will be provided as supporting documents to demonstrate the extent and the geographical coverage of the PAIs within the GP. All the KML files will indicate the regions within the geographical boundaries of the host country Rwanda.

1.14 Conditions Prior to Project Initiation

The baseline scenario is the same as the conditions existing prior to the Grouped Project initiation. The majority of people in the host country still use traditional cookstoves or open fires which consume a large amount of fuelwood. The project aims to reduce the environmental and social burdens caused by inefficient biomass cooking by introducing energy efficient cookstoves to the individual households. The current scenario as established is continuation to use traditional wood fuel- based stoves to meet thermal energy demand for cooking.

For further information regarding this, please refer Section 3.4 (Baseline Scenario) below.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

A review of the national energy policies of the Rwanda shows that there are no mandatory laws, policies or requirements mandating the use of ICS. Though the Government of Rwanda's (GoR) Energy Group Strategic Plan (2019-2024) identifies dissemination of improved cookstoves (ICS) as one way to reduce fuel wood consumption. The government promotes the use of improved ICS, however it's not mandatory.

The GP complies with applicable local laws and regulatory frameworks. Under Rwandan law and regulation, none of the activities or processes relating to the manufacture or distribution of Improved cookstoves require Environmental Impact Assessments not meet any of the screening criteria in Appendix 2, page 40 of the EIA guidelines[4]. The VCS grouped project is a regulatory surplus and is not mandated by any law within the host country. Further information on this may be accessed from the Section 3.5 within this document.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

If yes, provide the registration number and the date of project inactivity under the other GHG program.

Is the project active under the other program?

☐ Yes ☒ No

[4] General Guidelines and Procedure for Environmental Impact Assessment, Republic of Rwanda, November 2006 (See filename: RW_EIA_Guidelines_Final_versionl_Nov_2006)

Project proponents, or their authorized representative, must attest that the project is no longer active in the other GHG program in the Registration Representation.

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

If yes, provide the program name(s), reason(s) and date for the rejection, justification of eligibility under the VCS Program, and any other relevant information.

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, “Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities].”

☐ Yes ☒ No

If yes to all:

Provide evidence of the public statement. Evidence must be provided in this section or in an appendix.

1.18 Sustainable Development Contributions

The GP is contributing towards sustainability, which is also verifiable through the SDG indicators selected for the specific activities that are going to be monitored for their fulfilment.

Following sustainable contributions were part of the PAI – 1 implementation (Distribution of ICS in Rwanda):

<u>SDG-1: No Poverty</u>

Approach: For arriving at the indicator value, PP monitor the relevant parameter for this SDG. In this case, the questions specific to the access to cooking technologies that are available to the households (HHs) prior to the project ICS distribution is counted towards the estimation of the indicator to this SDG.

Relevant SDG Indicator: 1.4.1 Proportion of population living in households with access to basic services

Available Data: The parameter that will be monitored to calculate the number of HHs living without access to basic services will be the number of beneficiaries. This is because the number of HHs that are unable to purchase any improved cooking devices can be considered as the ones living without access to the basic services such as clean cooking. The list of HHs which are to be distributed with the project ICS is provided by the government of Rwanda and is based on the economic classification list of the population of Rwanda.

Monitoring Requirement(s): The new project ICS beneficiary/user will be considered as living without the access to clean cooking, and the number of people using the project ICS as their primary device will be considered towards the estimation of this SDG impact. Clean cooking is considered as a basic service, and access to it is an addition to the SDG impact.

SDG-3: Good Health and Well Being:

Approach: For arriving at the indicator value, PP monitor the relevant parameter for this SDG, specifically the SDG indicator. This SDG relates to health, and DelAgua's project ICS is improving health for every individual who is receiving it.

Relevant SDG Indicator: 3.9.1 Mortality rate attributed to household and ambient air pollution

Available Data: Regarding indicator 3.9.1, at every project ICS beneficiary HHs, ones who are experiencing any improvement (or no improvement or deterioration) in their ambient air quality as compared to pre-project scenario will be considered as an input.

Monitoring Requirement(s): The data will be collected at the time of monitoring survey. Beneficiary HHs for VCS GP will be sampled for monitoring as per monitoring plan, and these are then requested to respond to survey questions, which is then recorded within the monitoring survey database. The percentage of people experiencing an improvement in air quality (i.e. decrease in smoke levels in HH due to cooking) will count towards a reduction in the mortality rate attributed to household and ambient air pollution. The number of people will be equal to the product of average household size identified during monitoring surveys and the number of total HHs received the ICS and the proportion of HHs which respond that they've experienced a decrease in smoke levels from the baseline (in response to the survey question).

SDG-4: Quality Education

Approach: Here, each household is trained for sustainable practices, best usage practices for ICS usage, and utilization of fuel resources. Apart from this, the project being one of its kind, provides education and training to 5000+ Community Health Workers (CHWs) engaged with us for distribution and end-user engagement.

Relevant SDG Indicator: 4.7.1 Extent to which (i) global citizenship education and (ii) education for sustainable development are mainstreamed in (a) national education policies; (b) curricula; (c) teacher education; and (d) student assessment

Available Data:

(i) Number of project ICS beneficiary HHs who will be visited by DelAgua staff for HH visit surveys will be used to calculate the parameter. This data is continuously recorded in the distribution master database.

(ii) Number of CHWs that will be trained for the project activity implementation (distribution and surveys).

Monitoring Requirement(s): The distribution master database and the CHW training data will be used to calculate the number of people trained under the project activity.

SDG-5: Gender Equality

Approach: In this case, the number of ICS beneficiary HHs will be questioned over their time spent for collecting fuel (firewood).

Relevant SDG Indicator: 5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location

Available Data: Monitoring Survey for VCS GP includes question on time spent on collection of fuel.

Monitoring Requirement(s): The value will be recorded in the monitoring surveys, and then saved in the survey database. The total number of HHs which acknowledge that the amount of time spent on collection of firewood has changed will be arrived by calculation of survey data – i.e. multiplying the proportion of survey respondents accepting a decrease in time spent on cooking fuels with the total number of households.

SDG-7: Affordable and Clean Energy

Approach: For arriving at the indicator value, PP monitor the relevant parameter for this SDG, specifically the SDG indicator.

Relevant SDG Indicator: 7.1.2 Proportion of population with primary reliance on clean fuels and technology

Available Data: Distribution Master Database includes the data for this, where each HH where project ICS is distributed, acts as an input. The data within the database is updated on a continuous basis.

Monitoring Requirement(s): Distribution Database is exported for the monitoring period. This gives the number of beneficiaries who are now benefiting from the clean and affordable energy for the Monitoring Period.

SDG-8: Decent Work and Economic Growth

Approach: For arriving at the indicator value, PP monitor the relevant parameter for this SDG, specifically the SDG indicator.

Relevant SDG Indicator: 8.3.1 Proportion of informal employment in total employment, by sector and sex

Available Data: The salary slips for the CHWs act as the data source since these are not on DelAgua's payrolls. However, there are different data sources (number of mobile phones assigned to individuals during distribution, number of CHWs trained for a specific distribution event), which are used to total the CHWs working for DelAgua.

Monitoring Requirement(s): This data is continuously collected and is obtained from Training and Account teams.

SDG-13: Climate Action

Approach: For this SDG, there are no separate monitoring requirements. PP declare the GHG Emission Reductions achieved by project in the monitoring period. This is verified by a VCS accredited VVB, and the resultant VCUs are then issued, acting as the net impact contributed towards this SDG.

Relevant SDG Indicator: Not applicable.

Available Data: This can be accessed from the Monitoring Reports that are publicly available.

Monitoring Requirement(s): All the monitoring requirements are already detailed in the Monitoring Reports and corresponding ER Sheets.

SDG-15: Life on Land

Approach: For arriving at the indicator value, PP monitor the relevant parameter for this SDG, specifically the SDG indicator.

Relevant SDG Indicator: 15.4.2 Mountain Green Cover Index

Available Data: (i) Baseline (pre-project) survey data, (ii) Monitoring Surveys (periodic), (ii) Registered PD&MR and ER Sheet

Monitoring Requirement(s): This parameter is fixed for per ICS (HH level) firewood consumption, which is then multiplied with monitored values (such as usage rate), other relevant values as per the equations provided in the methodology applied for VCS GP. Based on the calculations, per HH level firewood savings are calculated.

This is then multiplied with the total number of ICS distributed. It gives the total amount of firewood saved by the PAI (or GP) in the specified monitoring period. This will be the amount of firewood that is saved from being harvested from the valuable mountain forest ecosystem.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

Not applicable as the project adopts a net gross adjustment factor of 95% to account for leakage as per applied methodology .

1.19.2 Commercially Sensitive Information

All information in this Grouped Project Description is public.

1.19.3 Further Information

Not Applicable.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	The primary stakeholders for the project are the recipients of the improved cookstoves i.e. rural households across Rwanda. Other stakeholders include project staff employed by DelAgua, the Ministry of Environment, other govt. bodies, relevant NGOs and other organizations which are responsible for promoting clean cooking within Rwanda.
Legal or customary tenure/access rights	As the project is not a land use project, there are no legal or customary tenure issues or access rights that are impacted by the project.
Stakeholder diversity and changes over time	The primary stakeholders in the project are rural households across Rwanda. No expected change has happened to the stakeholders' makeup.
Expected changes in well-being	Household air pollution (HAP) is estimated to lead to 3.2 million[5] premature deaths globally each year, with the majority of those occurring in sub-Saharan Africa and Asia[6]. By providing improved

[5] https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health

[6] https://ourworldindata.org/grapher/indoor-pollution-deaths-1990-2017

	cookstoves and education on the benefits of clean cooking, the project will see positive changes in health. In addition, clean cooking has been shown to contribute towards various SDGs, such as gender equality and zero hunger, meaning the project will produce positive changes in well-being beyond health.
Location of stakeholders	Primary stakeholders are located in rural communities across Rwanda. District government officials are located in rural areas across Rwanda, whilst the Ministry of Environment is headquartered in Kigali (capital city.)
Location of resources	As this is not a land use project, this is not applicable.

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	01-March-2023
Stakeholder engagement process	The Local Stakeholder meeting, under VCS Standard v4.4 with NGOs, Academia, other project developers, and any other entity who concerns itself with the DelAgua clean cookstove distribution was conducted at national level on 01-March-2023. The relevant stakeholders for the LSC were invited using various mode to get the participation/feedback which included publication in newspaper "The New Times" dated 03 & 13 February 2023, email sent to identified NGOs and other organizations dated 04 & 13 February 2023 and advertisement in radio channels. The local stakeholder consultation meeting was held on 01-March-2023 from 9.00 AM to 11.00 AM at Gorillas Golf Hotel, Nyarutarama, Rwanda. 

	 **Additional Stakeholder Engagements** In addition, the PP conducted numerous rural stakeholder engagements prior to the project cookstove distributions at a District and Sector level in July 2022. These meetings were intended to do the following for the rural communities who would be the ultimate beneficiaries of the stoves: 1. To gather local people together and inform them about the VCS grouped project activity. 2. To make them aware about the Carbon Finance without which this VCS grouped project could not be established. 3. To make the local people understand the benefits of clean cooking. 4. To demonstrate the working of sample cookstove that is aimed for the distribution. 5. To collect their feedback, queries, concerns, and respond those at the meeting itself. 6. Provide updates to community members of when the stoves will be arriving ready for disbursement and education. The LSC meetings were recorded using digital methods (audio-visual, photographic, and pdf records of the Q&A along with attendance), and paper format of the meeting forms. The invites were also shared with the local stakeholders a month before the meeting date. The evidence were shared with the VVB during the validation and has been validated.
Consultation outcome	The agenda of the meeting was to showcase the VCS grouped project being implemented and invite local stakeholders to understand about the project, submit their grievances, and raise

	their concerns, wherever required. Below are the steps followed during stakeholder consultation meeting 1. The meeting started with presentation on grouped project activity and associated benefits to society, ecology and economic aspect due to use of improved cookstoves. 2. The sample cookstove units, which were being distributed under VCS 3699 implementation, was demonstrated to the local stakeholders present for its usage. 3. During the meeting, stakeholders were invited to raise their queries and concerns. These were duly responded to during the meeting itself.
Ongoing communication	As far as continuous grievance of the local stakeholders is concerned, the required measures are already in place to account for a grievance through any media. Following are the channels through which the local stakeholder can communicate with the PP during the implementation of GP: 1. PP (DelAgua) has a toll-free helpline for the relevant stakeholders to connect with them in case they require any assistance. 2. The DelAgua CHW network is always ready to obtain any feedback from the end users and local authorities. Most of the feedback is received by PP through this medium, since a lot of people in rural Rwanda do not have access to phones. 3. PP distribution team informs the end users during distribution that they may write their feedback on paper, and submit it to the nearest cell office, and PP team will receive it through the government officials. These medium of interaction with local stakeholders will be verified by the VVB at the time of validation and shall remain open for Verra for review. In line with the VCS requirements, these documents shall be recorded and maintained for a minimum duration of 2 years post the end of the project implementation.
Stakeholder input	Below are the queries raised by the attendees Q.1. Whether current grouped project activity of DelAgua involves only improved cookstove distribution? Ans: The project activity involves only Improved cookstove distribution. Q.2. Which districts/region the project is planned under the current project activity? Ans: The PP representative explained that the project boundary for the grouped project activity is host country Rwanda and various

	districts will be targeted where ICS penetration is still less or not there. Q.3. Whether any changes in ICS models to be used under current project than ICS distributed under other ongoing projects as DelAgua group is already having other registered projects and distributing ICS from past many years. Ans: The technology upgrades with time and PP uses the state-of-the-art ICS technology available at the time of start of the grouped project activity, under current project the PP is distributing Burn make Kuniokoa generation 2 and SSM32 models which are having more than 50% thermal efficiency. There were no negative comment/concern raised by stakeholders attended the meeting.

2.1.3 Free Prior and Informed Consent

Obtaining consent	Prior to commencing the project, DelAgua signed a Memorandum of Understanding with Ministry of Environment to distribute improved cookstoves to rural households across Rwanda. Following this, DelAgua has signed Collaboration Agreements with the districts that the project has been delivered in (to date.) These agreements help the population in protecting the environment by scaling down firewood consumption while reducing indoor air pollution leading to improved respiratory health. Prior to distribution all potential ICS recipients are gathered to introduce the project. They are invited to receive a ICS free of charge. All recipients must consent to receive an ICS by attending a distribution day and signing the agreement on the bottom of the DelAgua poster.
Outcome of FPIC	Not applicable, the project activity involves distribution of improved cookstoves to individual households and hence any possibility of forced eviction/physical relocation of people due to the project is not applicable. There are no legal or customary tenure/access rights to territories and resources, including collective and conflicting rights, held by stakeholders, indigenous people (IPs), local communities (LCs), or customary rights holders. It is a completely voluntary activity and households participating are free to choose whether they take part or not. The PP collaborates with the local entrepreneurs for fabrication as well as installation of project ICS thereby empowering them economically rather than relocation / threat.

2.1.4 Grievance Redress Procedure

Development process	The grievance redress procedure was developed to ensure that local stakeholders have accessible and culturally appropriate channels to raise concerns. The process includes receiving, hearing, responding to, and resolving grievances within a reasonable time frame. Various communication methods have been established to accommodate different stakeholder needs, especially considering rural communities with limited phone access as stated below.
Grievance redress procedure	To address continuous grievances from local stakeholders, measures have been put in place to allow grievances to be reported through multiple channels: 2. **Toll-Free Helpline** – DelAgua provides a toll-free helpline for stakeholders to connect and seek assistance. 3. **Community Health Worker (CHW) Network** – The CHW network collects feedback directly from end users and local authorities. This method is widely used as many rural Rwandans lack phone access. 4. **Written Submissions via Local Government Offices** – During stove distribution, end users are informed that they can submit written feedback to the nearest cell office, where it will be relayed to DelAgua through government officials. These mechanisms ensure that grievances are received, documented, and addressed in a structured and transparent manner.

2.1.5 Public Comments

Comments received	Actions taken
No comments have been received during the public commenting period.	N/A

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

DelAgua has been delivering cookstove projects for over 10-years and has several active projects listed under VERRA. DelAgua does not sub-contract any of the implementation of this project to non-DelAgua entities. Thus, DelAgua management teams have expertise or experience in implementing similar project activities and engaging communities.

The operational management team for this project are Rwandan nationals with deep experience of rural delivery of services and government engagement.

Senior management of DelAgua manage multiple projects across multiple jurisdictions with deep experience in Carbon Methodologies, Research, Data Analysis and management of large-scale development projects. DelAgua staff are well-trained, with prior experience in distribution and monitoring. They received training on the questionnaire before conducting the surveys.

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	Natural Threat: The climate crisis and natural disasters can precipitate displacement and migration, resulting in the relocation of households.	The grouped project involves distribution of ICS that are portable and lightweight. Consequently, users of the project's ICS can easily carry them in case relocation becomes necessary.
Risks to stakeholder participation	N/A	N/A
Working conditions	DelAgua staff at risk from working in remote locations	DelAgua maintains a Health, Safety & Security policy for all staff in Rwanda. In addition, DelAgua maintains a risk register (as part of an overall risk management policy) that tracks all risk types (in addition to safety and security.) This is updated on a quarterly basis.
Safety of women and girls	Household visits result in Project representatives visiting households to conduct surveys.	DelAgua ensures all staff follow a clear Code of Conduct that they sign up to upon recruitment. This policy is regularly updated and enforced. Toll Free number exists within Rwanda and a clear whistleblowing policy is adhered to for any grievances raised. Training includes modules on safety and security. All surveys must first gain consent of the respondent. This individual must be above 18 or the survey is immediately ended.

Safety of minority and marginalized groups, including children	Children are recruited to deliver the project.	DelAgua maintains a policy on child labour which all staff, suppliers and vendors are required to adhere to. This policy is regularly updated and enforced.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	No risk identified	N/A, the project does not involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous materials.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified[7]	Mitigation or preventative measure(s) taken
Discrimination	Risk of discrimination based on gender, race, religion, sexual orientation or other habits	DelAgua maintains several HR policies for staff as well as a safeguarding policy for beneficiaries. These policies are maintained, monitored and implemented by DelAgua's HR unit.
Sexual harassment	Risk of sexual harassment	DelAgua maintains several HR policies for staff as well as a safeguarding policy for beneficiaries. These policies are maintained, monitored and implemented by DelAgua's HR unit.
Equal pay for equal work	Risk of Potential disparity in pay for equal work.	DelAgua ensures fair compensation policies that align with the principle of equal pay for equal work, regularly reviewing

[7] The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

		salaries to prevent gender-based wage gaps.
Gender equity in labor and work	Risk of gender inequalities in recruitment process	DelAgua maintains recruitment policies that specifically commit the organization towards gender equity. Where possible, DelAgua works to try to increase female recruitment into the organization.
Forced labor	Risk of illegal act of recruiting, transporting, harboring, or receiving people through force, fraud.	DelAgua maintains policies on forced labour and requires that all staff, suppliers and vendors agree to adhere to these policies.
Child labor	Children are recruited to deliver the project.	DelAgua maintains policies on child labour and requires that all staff, suppliers and vendors agree to adhere to these policies.
Human trafficking	Risk of illegal act of recruiting, transporting, harboring, or receiving people through force, fraud.	DelAgua maintains policies on forced and child labour and requires that all staff, suppliers and vendors agree to adhere to these policies..

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	The project involves the distribution of improved cookstoves to individual households and is not a land use project. Therefore, in this context, there are no legal or customary tenure/access rights to territories and resources, including collective and conflicting rights held by stakeholders, indigenous people (IPs), local communities (LCs), or customary rights holders. The primary stakeholders are given the opportunity to provide their details if they wish to receive a free improved cookstove. It is entirely their choice whether to sign up or to decline. Once they have signed up, stoves are distributed irrespective of tribal, ethnic, religious or political backgrounds of the beneficiaries. Beneficiaries are not subjected to

	forced or coerced to use the stoves during ongoing monitoring work.

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken
No risk identified	N/A. The project involves distribution of improved cookstoves to households in the Rwanda and does not affect the cultural heritage sites. Primary stakeholders are invited to give their name during the beneficiary data collection progress if they wish to receive an improved cookstove. They are not compelled to do so, nor are they told to change their traditional practices beyond being encouraged to use their improved cookstove for future cooking events.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No risk identified	N/A. The project does not impact any IP, LC or any other customary rights involving resources, territories or properties.

2.3.5 Benefit Sharing

This section is not applicable, as the project does not impact the property rights of individuals.

Process used to design the benefit sharing plan	N/A
Summary of the benefit sharing plan	N/A
Approval and dissemination of benefit sharing plan	N/A

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk identified	The ICS reduces GHG emissions through its better design and improved performance as

		compared to traditional stove, without changing the baseline fuel or introducing additional pollutants. Therefore, the project has no impact on biodiversity and ecosystems.
Soil degradation and soil erosion	No risk identified	The project stoves directly decrease wood fuel demand and reduce the pressure from wood cutting within the project boundary. Thus, project activity will indirectly avoid erosion associated with tree cutting/ felling
Water consumption and stress	No risk identified	The project activity which is the distribution of improved cookstoves in the Rwandan households, does not affect water consumption patterns in the localities. Therefore, no risk identified.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

Does not require further assessment, as the project is not located in or adjacent to habitats for rare, threatened, or endangered species.

2.4.2 Introduction of Species

This section is not applicable to this grouped project as it does not involve planting and/or species introduction.

2.4.3 Ecosystem Conversion

This section is not applicable, as the grouped project does not belong to the categories of ARR, ALM, WRC or ACoGS.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0050	VM0050 - Energy efficiency and fuel-switch measures in cookstoves	1.0
Tool	CDM TOOL30[8]	Calculation of the fraction of non-renewable biomass	4.0
Guideline	-	Guideline: Sampling and surveys for CDM project activities and programmes of activities	4.0
Standard	-	Standard: Sampling and surveys for CDM project activities and programmes of activities	9.0

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
VM0050 v 1.0	The project activity corresponds to: a) Replacement of non-renewable biomass (e.g., firewood, charcoal)-fired cookstoves with any of the following: i) More efficient project devices that use the same fuel as in the baseline; ii) Efficient project devices fired by renewable biomass or bioethanol; iv) Electric-powered project devices. b) Replacement of solid or liquid fossil fuel (e.g., coal, kerosene)-fired cookstoves with any of the following:	The project activity involves the distribution of improved wood-fuel-based cookstoves (ICS) to replace traditional three-stone fire stoves in beneficiary households and communities. Thus, the project meets the applicability conditions, as it distributes more efficient project devices that use the same fuel as in the baseline.

[8] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-30-v4.0.pdf

	i) Efficient project devices fired by renewable biomass or bioethanol; ii) Efficient project devices fired by LPG; or iii) Electric-powered project devices.	
	Project devices are used in households, communities, institutions, or small or medium enterprises (SMEs), collectively referred to in this methodology as the "target population	The project activity involves the distribution of improved wood-fuel-based cookstoves (ICS) to replace traditional three-stone fire stoves in beneficiary households and communities. Thus, the project meets the applicability conditions
	Where renewable biomass is used, it is exclusively renewable and qualifies as one of the following: a) A by-product, residue, or waste stream from agriculture, forestry, and related industries; or b) Originating from dedicated plantations that comply with all relevant applicability conditions in the most recent version of CDM TOOL16	The grouped project activity does not involve fuel switch measures. Hence criterion not applicable.
	Where biomass residues are used, they would have been left to decay or burned without energy recovery before implementation of the project activity, and their use does not involve a decrease in carbon pools – in particular of dead wood, litter, or soil organic carbon – on the land	Not Applicable

	areas from which the biomass residues originate.	
	Where biomass residues from a production process are used, project implementation does not result in an increase in the processing capacity of raw input or any other substantial changes (e.g., product change) in this process.	Not Applicable
	Where more than one type of renewable biomass is used, each of the biomass types used complies with the applicability conditions.	Not Applicable
	Where project activities introduce renewable biomass as charcoal, it is renewable charcoal produced by efficient charcoal production processes (e.g., retort sedentary kilns, improved sedentary kilns, Casamance kilns). Methane produced during the charcoaling process is captured and destroyed or combusted for energy purposes.	Not Applicable. The project activity involves the distribution of improved wood-fuel-based cookstoves (ICS) to replace traditional three-stone fire stoves in beneficiary households and communities.
	Project devices using renewable biomass (fuel-switch) or non-renewable biomass (improved efficiency) are single-pot, multi-pot portable, or in-situ cookstoves with an initial thermal efficiency of at least 25%.	The ICS distributed in each PAI has a thermal efficiency of at least 25%. Refer to Section 1.12 for details on the example ICS distributed or planned for distribution under the project, along with their specifications.
	Project devices using LPG or bioethanol are single-pot, multi-pot portable, or in-situ	Not Applicable

	cookstoves with an initial thermal efficiency of at least 30%.	
	Electric project devices meet the maximum risk factor score of on the Cookstove Durability Protocol and have the following minimum thermal efficiency: a) Hot plates and electric hobs: 40% b) Induction stoves and other electric stoves: 70%	Not Applicable
	Project devices using LPG comply with all of the following conditions: a) The baseline fuel either includes non-renewable biomass or is a more carbon-intensive fossil fuel (demonstrated by the baseline survey); b) The project has a provision for metering LPG supplied to each consumer at the LPG filling station, in order to determine household LPG consumption; and c) The project does not seek to issue any carbon credits for periods after 31 December 2045.	Not Applicable
	Electric project devices use the following electricity sources: a) Decentralized renewable energy systems: Decentralized energy systems using fossil fuels are not eligible, except for backup generators that supply less	Not Applicable

	than 1% of the annual electricity of the decentralized renewable energy system. b) Self-generated renewable electricity (with a maximum of 20% electricity from non-renewable sources for backup); or c) National or regional electricity grid.	
	The project proponent designs incentive mechanisms to reduce the use of inefficient baseline devices and practices that can be replaced by the project devices and describes these mechanisms in the project description.	One of the primary reasons households, especially in rural areas, continue to use inefficient baseline stoves is affordability. To address this, the PP distributes ICS free of cost, ensuring that households can access and adopt these improved stoves without financial barriers. This initiative directly facilitates the replacement of inefficient stoves. Additionally, the PP provides awareness programs for beneficiaries, educating them on the proper use and benefits of the project stoves. By fostering user understanding and engagement, these efforts further encourage the transition away from inefficient baseline stoves, promoting sustained adoption of the improved cookstoves.
	Where a project device has ended its technical life, the project proponent either	The PP has the provision that, upon the end of the ICS's lifetime, the device will either

	replaces it with a comparable or better project device or retrofits its essential components to continue meeting the minimum service level requirements (i.e., thermal energy generation), otherwise no further emission reductions may be claimed for the project device.	be replaced with the same or a comparable or better project device or have its essential components retrofitted to continue meeting the minimum service level requirements (i.e., thermal energy generation). Otherwise, no further emission reductions may be claimed for the project device.
	Project proponents implement a method for the distribution and identification of project devices that avoids double counting of emission reductions by other mitigation actions and includes unique product identification on the stove itself at the time of distribution/sale (e.g., program logo, alpha/numeric ID, and end-user location, such as geographic coordinates, complete address information).	To prevent double counting of emission reductions, each cookstove is assigned a unique identifier number, and its distribution is recorded in a database containing specific details for each recipient. Each stove is equipped with a unique barcode, which, when scanned, provides access to all end-user information associated with that ICS, ensuring transparency and eliminating the risk of double counting. Each PAI implements a system to prevent double counting, with data collected through a smartphone app where possible or via paper forms when necessary. The recorded information includes: • Date of distribution • Unique serial number for each stove • End-user details such as name, phone number, address/physical location, and ID number.
	The project complies with any national, sub-national, or local regulations or guidance	The project complies with national, sub-national, and local regulations and guidance for the installation, commercialization,

	for the installation, commercialization, distribution, and use of improved cookstoves and/or fuel supply and use for the target population. National, regional, and local regulatory frameworks for the provision of the type of thermal energy services provided by the project activity must be documented.	distribution, and use of improved cookstoves and fuel supply for the target population. In Rwanda, the implementation of improved cookstove projects aligns with national policies aimed at promoting clean cooking solutions and reducing reliance on traditional biomass fuels. The Government of Rwanda has introduced strategies such as the Biomass Energy Strategy (2019-2030) and updates to its Nationally Determined Contributions (NDCs) to encourage the adoption of efficient cookstoves.
	Where project activities reduce emissions from non-renewable biomass, including firewood and charcoal, the risk of double counting is assessed on a national basis by evaluating at validation and crediting period renewal whether there are REDD+ projects or jurisdictional REDD+ programs whose project boundary overlaps with the expected fuel source area of the project. The project proponent must report on the findings of this assessment for informational purposes in the project description.	There are no REDD+ projects in the host country. Therefore, the risk of double counting from overlapping project boundaries does not apply.
CDM TOOL30 v.04	This tool may be used by: (a) DNAs to submit region- or country-specific default f_{NRB} values, following the procedures for development, revision, clarification and	*Option (b) applied to calculate the project specific f_{NRB} value as described under in the separate f_{NRB} spreadsheet submitted to VVB.*

	update of standardized baselines (SB procedures); or (b) project participants to calculate project- or PoA-specific fNRB values.	
	Project participants and DNAs shall identify and clearly delineate the applicable area for which fNRB is determined. For project- or PoA-specific fNRB values, project participants shall assess and use the area from which woody biomass is sourced for end-users included in the project activity and justify the selection of the area in CDM project design documents. For region- or country-specific fNRB values, DNAs shall specify the applicable area.	*The PP has chosen host country Rwanda as region for the project activity.*
	Project participants and DNAs may choose between the following two options to determine the value of fNRB: (a) Use the default value as provided in TOOL33; or (b) Calculate fNRB by determining the share of renewable and non-renewable woody biomass in the total quantity of woody biomass consumption for the country/region or the project area (hereinafter referred as the applicable area) following the procedure and	The PP has chosen option b), wherein a third-party consultant has established the f_{NRB} for host country. *Please refer Appendix-1 for calculation details and comparison with literature.*

	requirements in the paragraphs below. The project participants shall compare and analyse the calculated values against the values for fNRB reported in relevant scientific literature and justify any differences. This analysis shall be included in the appropriate section of the PDD. The relevant scientific literature should include at least: (i) Bailis, R.; Drigo, R.; Ghilardi, A. & Masera, O. (2015). The carbon footprint of traditional woodfuels. Nature Climate Change, 5(3), pp. 266–272.	
	If the f_{NRB} value is estimated at the national level, as a cross check, project proponent shall compare the value of estimated NRB with the product of: i) total average above ground biomass tonnage of the area of forest areas deforested in recent past (tonnes/ha), and ii) most recent available observed annual rate of deforestation (ha/yr). If the estimated NRB value is more than 10% above the value calculated as per the product of biomass and deforestation rate, justification shall be provided for the higher value for NRB	Please refer Appendix-1 for cross check.

3.3 Project Boundary

The Grouped Project boundary is defined as per the applied methodology VM0050, ver. 1.0.

Table 7: The GHG sources & sinks within the project boundary

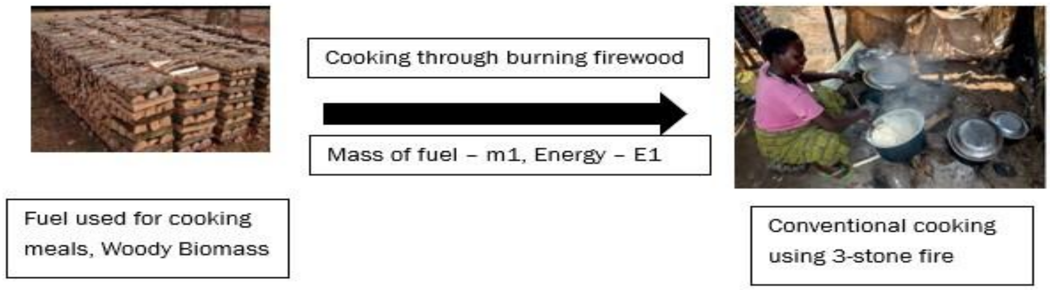
Source		Gas	Included?	Justification/Explanation
Baseline	Source 1– Thermal energy generation (Emission from burning of non-renewable Biomass, i.e. Woody Biomass for household cooking requirements)	CO_2	Yes	Major source
		CH_4	Yes	Major source
		N_2O	Yes	Major source
		Other	No	No other source identified
Project	Source 1– Thermal energy generation (Emission from burning of non-renewable Biomass, i.e. Woody Biomass for household cooking requirements)	CO_2	Yes	Major source
		CH_4	Yes	Major source
		N_2O	Yes	Major source
		Other	No	Not applicable

Figure 4: Project boundary map showing physical locations of the Grouped Project distribution

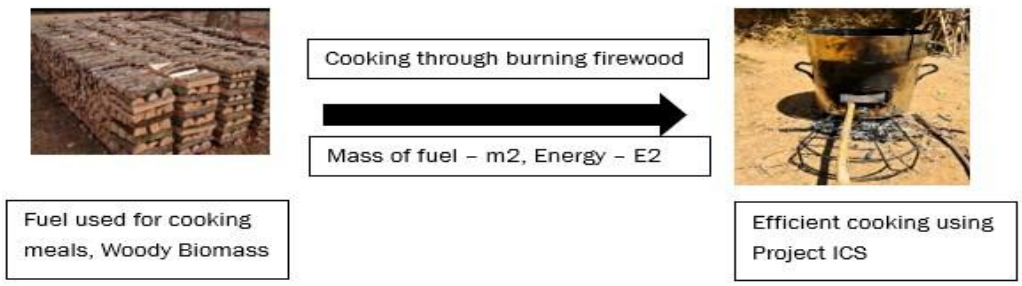
As per the template requirements, the diagram of the equipments distributed under the grouped project is provided in Section 1.

Flow of mass and energy within the grouped project activity is described below.

Baseline Scenario:



Project Scenario:



Here, the change in mass of firewood between baseline and project scenario is m1>m2.

The change in energy consumed for cooking between baseline and project scenario is E1<E2.

The systems that were applied for the distribution activity to take place, are described in the Section 5.3 of this document.

3.4 Baseline Scenario

For this project activity, the baseline scenario represents the continued use of traditional wood-fuel cookstoves by rural households across Rwanda, where domestic users rely on inefficient cooking technologies due to limited access to improved alternatives. In the absence of the project, these households would persist in using solid biomass as their primary fuel source, leading to high fuel consumption, increased deforestation, and elevated greenhouse gas emissions. The inefficiencies of traditional stoves contribute to significant environmental and health impacts, reinforcing the need for intervention.

1. Selection and Justification of the Baseline Scenario

Step 1: Alternative baseline scenarios:

As per the Fifth Rwanda Population and Housing Census, 2022 thematic report on Characteristics of Households and Housing[9] , prepared by the Ministry of Finance and Economic Planning and the National Institute of Statistics of Rwanda, and released in July 2023, more than 93% of total households in Rwanda rely on solid biomass as a cooking fuel. Among rural households—who are the primary beneficiaries of the project—over 97% depend on biomass. Specifically, 93.3% of rural households rely entirely on firewood and 4% on the charcoal as their baseline fuel, while less than 1% of the rural population uses clean fuel for cooking.

Table 8.14: Distribution (count and %) of the private households by main source of energy for cooking by province and area of residence

Province and Main source of energy for cooking	Count			Percentage		
	Urban	Rural	Total	Urban	Rural	Total
Rwanda						
Firewood	329,422	2,190,867	2,520,289	34.2%	93.3%	76.1%
Charcoal	478,164	94,984	573,148	49.6%	4.0%	17.3%
Gas	129,507	21,265	150,772	13.4%	0.9%	4.6%
Electricity	803	190	993	0.1%	0.0%	0.0%
Kerosene/Parafine	155	87	242	0.0%	0.0%	0.0%
Biogas	138	149	287	0.0%	0.0%	0.0%
Solar Power	76	589	665	0.0%	0.0%	0.0%
Crop waste	2,884	14,272	17,156	0.3%	0.6%	0.5%
Animal dung	34	270	304	0.0%	0.0%	0.0%
Briquette	42	285	327	0.0%	0.0%	0.0%
Peat	10	27	37	0.0%	0.0%	0.0%
Sawdust	245	723	968	0.0%	0.0%	0.0%
Other cooking fuel	249	1,031	1,280	0.0%	0.0%	0.0%
Never cook	22,513	23,634	46,147	2.3%	1.0%	1.4%
Not Stated	45	83	128	0.0%	0.0%	0.0%
Total	964,287	2,348,456	3,312,743	100.0%	100.0%	100.0%

Figure 1 Table 8.14: Fifth Rwanda Population and Housing Census, 2022

Also, as per the latest available DHS country data, 98.9%[10] of rural populations use some type of solid fuel for cooking, with 71% rural household's dependent on firewood and 20.9% use straw/shrubs/grass for cooking. The same reports states that only 1% of rural population and 3.6% of total Rwanda population use clean fuel for cooking. This is evidenced by the latest available Demographic and Health Survey Report (DHS) 2019-20, published in September 2021 by the National Institute of Statistics of Rwanda in collaboration with the Ministry of Health (MoH), supported by funding by the Government of Rwanda and various multilateral organizations.

[9] https://statistics.gov.rw/file/14187/download?token=dCyOX3ty

[10] https://dhsprogram.com/publications/publication-FR370-DHS-Final-Reports.cfm

Table 2.3 Household characteristics

Percent distribution of households and de jure population by housing characteristics, percentage using solid fuel for cooking, percentage using clean fuel for cooking, and percent distribution by frequency of smoking in the home, according to residence, Rwanda DHS 2019-20

	Households			Population		
Housing characteristic	Urban	Rural	Total	Urban	Rural	Total
Electricity						
Yes	86.4	36.7	45.7	86.4	38.4	46.6
No	13.6	63.3	54.3	13.6	61.6	53.4
Total	100.0	100.0	100.0	100.0	100.0	100.0
Flooring material						
Earth/sand	20.2	75.3	65.3	20.3	74.5	65.2
Dung	0.0	0.3	0.2	0.0	0.3	0.2
Bricks without cement	0.6	1.4	1.2	0.7	1.4	1.3
Parquet or polished wood	0.0	0.0	0.0	0.0	0.0	0.0
Vinyl or asphalt strips	0.0	0.0	0.0	0.0	0.0	0.0
Ceramic tiles/coastal brick	11.2	0.7	2.6	13.5	0.8	3.0
Cement	67.8	22.4	30.6	65.3	23.0	30.2
Carpet	0.0	0.0	0.0	0.0	0.0	0.0
Other	0.1	0.0	0.0	0.1	0.0	0.0
Total	100.0	100.0	100.0	100.0	100.0	100.0
Rooms used for sleeping						
One	29.4	17.2	19.4	17.2	11.1	12.1
Two	30.2	40.4	38.5	30.1	38.4	37.0
Three or more	40.3	42.5	42.1	52.7	50.5	50.9
Total	100.0	100.0	100.0	100.0	100.0	100.0
Place for cooking						
In the house	24.1	21.9	22.3	20.6	19.5	19.7
In a separate building	42.6	61.6	58.2	49.6	65.0	62.3
Outdoors	32.5	16.3	19.2	29.6	15.5	17.9
No food cooked in household	0.8	0.2	0.3	0.2	0.1	0.1
Other	0.0	0.0	0.0	0.0	0.0	0.0
Total	100.0	100.0	100.0	100.0	100.0	100.0
Cooking fuel						
Electricity	0.5	0.1	0.2	0.5	0.1	0.2
LPG/natural gas/biogas	18.2	1.1	4.2	15.8	0.9	3.4
Kerosene	0.1	0.0	0.0	0.0	0.0	0.0
Charcoal	53.9	7.8	16.2	53.8	6.9	14.9
Wood	22.9	70.0	61.4	25.8	71.0	63.3
Straw/shrubs/grass	3.6	20.5	17.4	3.7	20.9	17.9
Agricultural crop	0.0	0.1	0.1	0.0	0.1	0.1
Animal dung	0.0	0.0	0.0	0.0	0.0	0.0
Briquette	0.0	0.0	0.0	0.0	0.0	0.0
Sawdust	0.1	0.0	0.0	0.1	0.0	0.0
Other	0.0	0.0	0.0	0.0	0.0	0.0
No food cooked in household	0.8	0.2	0.3	0.2	0.1	0.1
Total	100.0	100.0	100.0	100.0	100.0	100.0
Percentage using solid fuel for cooking[1]	80.3	98.5	95.2	83.3	98.9	96.2
Percentage using clean fuel for cooking[2]	18.7	1.3	4.5	16.3	1.0	3.6

Apart from the above, there are several other documentary evidences in the form of recently published reports and papers that can support the fact that the project's baseline was usage of firewood and unimproved cooking devices.

Some of these are listed below:

1. Ministry of Infrastructure: Biomass accounts for 85% of all energy consumed[11]. Under the National strategy for transformation, the sector objective is to halve the number of households using traditional cooking technologies to achieve a sustainable balance between supply and demand of biomass through promotion of most energy efficient technologies.

[11] https://www.mininfra.gov.rw/digital-transformation-1-1

2. Rwanda Integrated Household Living Conditions Survey, EICV5: In rural areas, firewood remains the most common type of cooking fuel, used by 93% of the households[12].

Thus, based on the above data, cooking using solid biomass fuel over traditional inefficient stoves is the prominent cooking practice. Since the Project Proponent (PP) provides improved cookstoves (ICS) exclusively to households that were previously using wood-based baseline stoves as their primary baseline technology, all selected beneficiaries meet the baseline requirement of relying on non-renewable wood fuel in traditional inefficient cookstoves.

PP conducted baseline survey and baseline KPTs between September to December 2021, to assess the type & quantity of fuels consumed and the cooking technologies in use in pre-project scenario. Given the above Surveys/KPTs were based on the older methodology, the PP has reconducted the baseline surveys (KPTs) in line with the applied methodology VM0050 ver 1.0 and the KPT protocol during January 2025. The baseline survey results confirmed that solid biomass fuel and solid biomass-based cooking technologies—primarily traditional stoves and three-stone fires—are the most widely used among households.

Therefore, the baseline scenario is the continued use of traditional cookstove such as three stone-fired cookstove using non-renewable wood fuel (firewood) by the target population in rural areas to meet similar thermal energy needs as provided by project cookstoves in absence of project activity. Further details on the baseline survey and Kitchen Performance Test (KPT) are provided in the following sections.

Based on the Government of Rwanda's reports and the baseline survey findings, it is confirmed that solid biomass is the primary baseline scenario. The only alternative scenario is the project activity without being registered under a GHG program[13].

Step 2: Existing and forthcoming government policies and legal requirements

Since it has been confirmed that solid biomass is the primary baseline scenario and the only alternative scenario is the project activity without being registered under a GHG program, the requirement to eliminate baseline alternatives that are inconsistent with existing or forthcoming mandatory legal or regulatory requirements is not applicable.

There are no government-imposed restrictions or binding regulations as per the policies reviewed14 that prohibit the continued use of traditional biomass cookstoves in the project region. Additionally, there are no mandatory efficiency standards or air quality regulations that would systematically enforce a transition away from traditional biomass-based cooking.

[12] https://www.statistics.gov.rw/publication/eicv-5-main-indicators-report-201617

[13] As per applied methodology page 12: The project activity without being registered as a project activity under a GHG program must be included as an alternative.

[14] Add 3-4 policies from 4151

Therefore, the consideration of alternative baseline scenarios remains unchanged, and no further eliminations are required under this criterion.

Step 3: Financial, institutional, and information barriers

As explained above, the only alternative scenario is the project activity without being registered under a GHG program. However, this scenario is not viable due to significant financial barriers.

Since the Project Proponent (PP) distributes improved cookstoves (ICS) free of cost to end users, the project relies solely on revenue from the sale of Verified Carbon Units (VCUs) to sustain its operations. Without registration under a GHG program, the project would lack a financial mechanism to cover the costs of cookstove production, distribution, and monitoring.

As a result, the alternative scenario of implementing the project without GHG registration is not financially feasible and cannot be considered a realistic baseline alternative. Therefore, no alternate baseline scenarios exist.

Thus, the remaining baseline scenario is the target population's continued use of inefficient cookstoves fired with non-renewable firewood to meet similar thermal energy outputs for cooking as the project devices. Therefore, the project activity remains consistent with the defined baseline scenario, ensuring compliance with the methodology.

2. Baseline Survey/KPT

PP conducted the baseline surveys (KPTs) in line with the applied methodology VM0050 ver 1.0 and the KPT protocol between 13 January 2025 to 17 January 2025 to assess the type & quantity of fuels consumed and the cooking technologies in use in pre-project scenario. The baseline survey was scheduled to ensure that no major festivals, holidays, or similar events were present, minimizing any potential impact on the outcomes.. The PP initially conducted a desk-research to identify regions (districts) where use of traditional cookstoves and wood fuel is predominant. This was followed by designing a survey questionnaire to validate the baseline scenario in the selected regions. According to these surveys, the baseline scenario was identified as the continued used of traditional cookstove without chimney or grate or three-stone fired for preparation of meal.

According to these baseline KPTs, all 128 sampled households (100%) were using a three-stone fire as their primary baseline stove.

The details of the sample size calculation for baseline surveys, sample selection criteria, procedures used for conducting KPT, and the results of KPT are as follows:

Sample Size Calculation:

The sample size was determined using CDM Guideline: Sampling and Survey for CDM PA and PoA version 4.0 and appendix 5 of the applied methodology. The sample size for parameter $BC_{b,I,y}$ (mean parameter) was calculated as 55 (each province wise is given below).

The PP ensured that calculated sample size met all appropriate methodological thresholds and is representative of the entire target population. Therefore, PP then oversampled to account for potential non-responses, data quality concerns, and hardware errors. This process resulted in a total target sample size of 150.

Sample Selection Criteria:

Rwanda has a total of 5 provinces – Eastern, Western, Northern, Southern, and Kigali. However, the 5th province, i.e. Kigali[15], includes 100% urban/peri-urban households and therefore is not included in the sampling procedure as these households lay outside the scope of the project.

The targeted sample size of 150 was divided among the four provinces of Rwanda - Eastern, Northern, Southern, and Western - resulting in approximately 35 samples per province. PP covered 12 rural districts across these four provinces (3 district from each province) of Rwanda as shown in the table below.

FINAL SELECTION			
Eastern Province	**Northern Province**	**Western Province**	**Southern Province**
District	District	District	District
Kirehe	Musanze	Nyamasheke	Nyamagabe
Ngoma	Gicumbi	Rusizi	Ruhango
Bugesera	Burera	Ngororero	Muhanga

These districts were generated randomly. Once a target of 35-40 samples per province was confirmed by the technical team, the operational team, who has previously received KPT data collection training from Berkeley Air Monitoring Group (BAMG), was instructed to target 7-8 villages per province and 5 samples per village. As per the literature review[16][17][18], cooking behaviour in the entire country is quite similar. Major dishes being cooked are beans, stew, rice, vegetable stew, porridge, leafy vegetable stew, Irish potatoes, and green banana. Considering similar cooking practices across the country, the selection of these villages was left up to the field team to account for logistical challenges throughout rural Rwanda. This was on the condition that the villages selected did not have any criteria that would make them non-generalizable to the wider population, and they were culturally, and socio-economically homogenous with the rest of the province. To identify households, a random walk was taken, where every 5th household, when walking in a random direction, was approached, and assessed against the protocol criteria. If they met them, they were invited to take part in the KPT.

Out of the 150 target samples, PP successfully conducted 128 KPTs in line with the KPT protocol. The sample coverage is as follows:

Province	Min. Sample Size Required	Sample Covered

[15] Administrative Structure of Govt. of Rwanda

[16] https://mecs.org.uk/wp-content/uploads/2022/04/Cooking-Diary-Study-Rwanda.pdf, table 6, page 20

[17] https://mecs.org.uk/wp-content/uploads/2022/02/MECS-EnDev-Rwanda-eCooking-Market-Assessment.pdf - page 8-9

[18] National Survey on Cooking Fuel Energy and Technologies in Households, Commercial and Public Institutions in Rwanda , Ministry of Infrastructure and Ministry of Finance, December 2020.

Eastern	17	33
Northern	10	33
Southern	14	33
Western	14	29
Total	55	128

Procedures followed for Conducting KPT:

The KPT surveys were conducted by a competent team trained for the KPT surveys by Berkeley Air Monitoring Group (BAMG) for performing KPTs, based on the KPT protocol. Once the villages were selected by the operational team, enumerators were issued with protocols for the selection of the individual households. These protocols outlined that the selected households should, but are not limited to, the following criteria in accordance with the KPT protocol:

- Have the person who is primarily responsible for cooking available for visits once a day for the next 4 days;
- Not have any events scheduled that would disrupt normal cooking practices during this time (e.g Weddings, community events);
- Primarily cook on wood

Additionally, the enumerators were instructed to follow, but are not limited to, the following criteria in accordance with the KPT protocol:

- Please avoid a measurement campaign during special occasions, holidays, or any day when cooking is not normal and be aware of local events like market days that may involve above-average fuel consumption. Be aware that 3 days of testing involves 4 days in contact with the family—the first day is spent briefing families, as explained in the next step.
- To explain the purpose of the test to the family members and arrange to measure their fuel consumption.
- Stress to household members that their cooking practices should remain as close to normal as possible for the duration of the test.
- Explain to family members that the surveyor will visit the respondent daily (for the next ~3 days) to collect the information.
- Ask the family to define an inventory area to store the fuel during the test. Record the weight and moisture content of the initial stock of fuels and if the family is going to collect or purchase fuel during the days of the test, ask them to keep newly collected or purchased fuel separate from fuel that has already been tested for moisture and weighed.
- Visit each household at roughly the same time each day, without being intrusive.

- Record fuel consumption by weighing the remaining fuel. Record the weight and moisture content of newly collected fuel before it is added to the family's stock.

- Moisture should be read the day before from the stock of fuel that is going to be used during the next 24-hour period. For each day, randomly draw 3 pieces of the fuel and measure its moisture in three positions.

PP also ensured conservatism in seasonal variation through adding a provision in the questionnaire in line with the KPT protocol[19]. These questionnaire for conducting baseline survey/KPTs were designed in line with Appendix 3:Binding Survey questionnaire of the applied methodology and the KPT protocol.

Baseline Survey and KPT Results:

A digitized data collection approach was taken by the team, where the survey responses and measurement records were recorded in the electronic survey forms through smartphones. The survey responses for each household were recorded in the centralized database, where all the survey results were combined, and the final KPT survey Excel sheet was exported.

The survey results from the baseline study are detailed as follows:

- **Baseline fuel type(s) and the percentage of their use by the target population:** According to the baseline survey conducted by PP, 124 out of 128 households reported using only wood fuel as their primary fuel, while 4 households indicated using both wood (primary) and charcoal (secondary) as fuel. This means that approximately 4% of households use both wood and charcoal, while 96% rely exclusively on wood for cooking. These findings closely align with the Fifth Rwanda Population and Housing Census, 2022[20] published by the Government of Rwanda, which reports that nearly 94% of the rural households depends on firewood as their primary cooking fuel and 4% on charcoal.
- **Source(s) of each baseline fuel and control households**: According to the baseline survey conducted by PP, 46 out of 128 households (~36%) reported purchasing fuel, while 82 out of 128 (~64%) stated that they collect fuel for cooking. These findings closely align with the 2020 National Survey on Cooking Fuel Energy and Technologies in Households, Commercial, and Public Institutions in Rwanda, which reports that nearly 69% of households purchase wood fuel. This confirms that these 128 households are statistically representative of the pre-project baseline conditions of the target population, consistent with the controlled household requirements on page 14 of VM0050 v1.0. Hence, the PP has allocated these 128 households as control households.
- **Baseline technologies for cooking and the percentage of their use by the target population**: According to the baseline survey conducted by PP, 124 out of 128 households reported using only three-stone fire, or a conventional cooking system as their primary baseline cooking technology, while 4 households indicated using both three-stone fire, or a conventional system and traditional charcoal stove as cooking technology. This means

[19] https://cleancooking.org/binary-data/DOCUMENT/file/000/000/604-1.pdf

[20] https://statistics.gov.rw/file/14187/download?token=dCyOX3ty

that approximately 4% of households use both, while 96% rely exclusively on three-stone fire, or a conventional cooking system for cooking.

- **Average household size (Hhi):** According to the baseline survey conducted by PP, the average reported household size in rural baseline households is approximately 4.31.
- The annual baseline fuel consumption per person was calculated from the KPT survey results of 128 households. The parameter was within the 95/10 confidence/precision which can be checked in the table below:

Kg/person/day assessment	**Baseline KPT**
Number of Samples	128
Median	2.39
Quartile (Q1)	1.69
Quartile (Q3)	3.38
IQR	1.69
Upper Quartile limit	5.91
Lower Quartile limit	0.00
95-10 Precision Assessment	**Baseline KPT**
Average consumption (kg/person/day)	2.57
Standard Deviation	1.14
Coefficient Of variance (CV)	44.44%
Lower Bound value of confidence interval	2.370
Upper Bound value of confidence interval	2.77
Confidence interval	0.40
Precision (2-sided)	7.70%
95-10 Rule Met?	**YES**
HH Size (Standard Adult Equivalent) (Hhi)	4.31
BCb,i,y (tons/person/yr)	**0.937**
BCb,i,y (tons/HH/yr)	**4.04**

The value obtained from the baseline KPTs is 0.937 tons per person per year. This result is scaled appropriately using the average household size (Hh_i) as follows:

$BC_{ex\text{-}ante,b,i}$ = $BC_{ex\text{-}ante,b,I}$ (tonnes/person/year) x Equivalent standard male adults (Hh_i) obtained from baseline KPT

= 0.937 x 4.31

= 4.04 tonnes/year

Comparison with Other Projects:

In order to benchmark results derived from the project baseline KPTs and to add another layer of assurance to their validity; the below table has consolidated baseline fuel consumption values for various registered projects across Rwanda:

Registry	**Project ID**	**Geography**	**Baseline fuel consumption value (tonnes/device/year)**	**Year of Baseline Assessment**
Verra	3654	Rwanda	6.753	2020
Verra	2380	Rwanda	7.635	2021

The value of fuel consumption derived using baseline KPTs by the project is conservative compared to values used by other VERRA registered projects which have conducted baseline assessment during the same time under VCS[21][22].

Comparison with Country Data:

The National Survey on Cooking Fuel Energy and Technologies in Households, Commercial, and Public Institutions in Rwanda, a report by the Ministry of Infrastructure and Finance (Government of Rwanda), indicates that the consumption of traditional biomass is approximately 6.75 tonnes per household per year. Therefore, the applied baseline fuel consumption value of 4.04 tonnes per household per year by PP is 40% lower than the government literature's value, making it the most conservative.

This comparison, aligned with registered projects and country data, ensures that the baseline fuel consumption value is conservative, in accordance with Section 2.2.1 of the VCS Standard.

Since the grouped project activity provides improved cookstoves to households where there was only firewood was used as the cooking fuel, and the cooking device was a traditional cooking device such as three-stone fired, traditional stove without chimney etc., it is considering the burning of firewood, and usage of unimproved cooking devices as the baseline. Also, PP verifies this for every individual household where distribution of project ICS is taking place, that they were using unimproved cooking device using firewood as fuel prior to receiving it. This is acknowledged by the end-user in the form of a signed ICS usage practices poster where the information is written down. This information can be checked through the sample posters. The poster is in local language Kinyarwanda, which makes it clearly understandable for the end user.

3.5 Additionality

The Additionality for Improved Cookstoves is demonstrated using the requirements provided in the methodology VM0050 version 1.0.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

[21] https://registry.verra.org/app/projectDetail/VCS/3654

[22] https://registry.verra.org/app/projectDetail/VCS/2380

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☐ No

If no, describe which mandated laws, statutes, or other regulatory frameworks require project activities and provide evidence of systematic non-enforcement to demonstrate regulatory surplus.

3.5.2 Additionality Methods

Step 1: Regulatory Surplus

There is no mandated government programme or policy in host country of this project ensuring the distribution of domestic fuel-efficient cookstoves. The project is not mandated by any law, statute or other regulatory framework, or for UNFCCC non-Annex I countries, any systematically enforced law, statute or other regulatory framework.

The Grouped Project is a voluntary activity initiated by the project proponent. The initiative is solely dependent on VCU revenue for its sustenance and is completely financed by the project developer

Step 2: Positive List

As per the Paragraph 7, step-2 of the methodology, the project activity must meet both of the following conditions to qualify for the positive list:

1) The project activity introduces:
   a) Efficient biomass-fired cookstoves that replace inefficient biomass-fired cookstoves;
   b) Efficient, solely renewable biomass-fired cookstoves that replace fossil fuel-fired cookstoves; or
   c) Electric cookstoves that replace inefficient biomass-fired or fossil fuel-fired cookstoves.
2) The project activity installs or distributes improved cookstoves at zero cost to the enduser and has no revenue source other than from the sale of verified carbon units (VCUs).

Projects that pass the regulatory surplus test (Step 1) and qualify for the positive list (Step 2) are deemed additional and are not required to apply Step 3.

<u>Justification</u>

In the case of this grouped project activity, all improved cookstoves being distributed are non-renewable, efficient biomass-fired cookstoves that replace inefficient biomass-fired cookstoves in Rwandan households. These ICS are distributed at zero cost to the end user.

Thus, the project complies with both Step 1 and Step 2, and therefore, the grouped project activity is deemed additional.

3.6 Methodology Deviations

Not applicable.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

As per the applied methodology VM0050, ver. 1.0, baseline emissions are calculated as follows:

$$BE_y = \Sigma EC_{i,y} \times N_{j,k,y} \times n_{j,k,y} \times (EF_{b,i,CO2} \times f_{NRB,y} + EF_{b,i,nonCO2}) \quad \text{Equation (1)}$$

Where:

BE_y	Baseline emissions during year y (t CO2e)
$EC_{i,y}$	Average energy consumption of baseline device type i in year y (TJ)
$N_{j,k,y}$	Number of commissioned project devices of type j from batch k in year y
$n_{j,k,y}$	Proportion of commissioned project devices of type j from batch k that remain operating in year y (fraction)
$EF_{b,i,CO2}$	CO_2 emission factor for fuel used by baseline device type i in the baseline scenario (t CO_2/TJ)
$f_{NRB,y}$	Fraction of woody biomass that is established to be non-renewable used by baseline device in year y (fraction)
$EF_{b,i,nonCO2}$	Non-CO_2 emission factor for fuel used by baseline device type i in the baseline scenario (t CO_2e/TJ)
i	Baseline device type and its respective fuel type
j	Project device type and its respective fuel type

The average energy consumption of baseline device type i is calculated as follows:

$$EC_{i,y} = BC_{b,i,y} \times NCV_{b,i} \quad \text{Equation (2)}$$

Where:

$BC_{b,i,y}$	Fuel used per baseline device type i during year y (tonnes)

$NCV_{b,i}$	Net calorific value of baseline fuel for baseline device type i (TJ/tonne)

Cross-check of $EC_{i,y}$ to address stove stacking:

Baseline consumption from the back calculation is calculated as follows:

$$EC_{est,y} = EC_{p,y} \times \eta_{new,avg,y} / \eta_{old,avg} \quad \text{Equation (3)}$$

Where:

$EC_{est,y}$	Back-calculated energy consumption of the potential mix of devices and fuels in the baseline in year y (TJ)
$EC_{p,y}$	Energy used in project scenario by project devices during year y (TJ)
$\eta_{new,avg,,y}$	Weighted average efficiency of project devices in year y (fraction)
$n_{old,avg}$	Weighted average efficiency of baseline devices that are replaced by project devices (fraction)

Note: Where the results indicate that baseline consumption ($EC_{i,y}$) is higher than that indicated by back-calculation from the project scenario ($EC_{est,y}$) then the back-calculation results ($EC_{est,y}$) will be applied in Equation (1) as a conservative cap.

$$ECp,,y = \sum_{j,k} BCp,j,k,y \times NCVp,j$$

Equation (4)

Where:

$BC_{p,j,k,y}$	Average quantity of fuel used by project device type j from batch k during year y (tonnes or m³)
$NCV_{p,i}$	Net calorific value of project fuel used in project device type I (TJ/tonne or TJ/m³)

Given that the grouped project involves the distribution of non-renewable biomass-based improved cookstoves exclusively to the target users, Equation (5) of the methodology applies to electric stoves and is therefore not applicable.

Sample Calculations of Estimated Baseline Emissions for the Year 2024 of CP1:

Parameter	Applied value/Approach
$BC_{b,i,y}$	= $BC_{b,i,y}$ (tons/person/year) * H_{hi} (average HH size) = 0.937 × 4.31 = 4.04 tons/year The values of $BC_{b,i,y}$ and Hh_i are derived from the Baseline KPT.
$NCV_{b,i}$	= 0.0156 TJ/tonne (Meth Default Value) = $NCV_{p,j}$

$EC_{i,y}$ $= BC_{b,i,y} \times NCV_{b,i}$ =4.04 × 0.0156 =0.063 TJ

$EC_{est,y}$ $= EC_{p,y} \times \eta_{new,avg,y} / \eta_{old,avg}$ = 0.019 × 0.4875 / 0.15 = 0.62 TJ

$\eta_{new,avg,y}$ = 0.4875 (assumed for ex-ante estimation only)

$\eta_{old,avg}$ = 0.15 (Meth Default Value)

Where $EC_{p,y}$ is calculated as follows:

$EC_{p,y}$ (tons/year) = $BC_{p,j,k,y}$ (tons/person/year) × $Hh_{j,k}$ × $NCV_{p,j}$

= 0.282 × 4.31 × 0.0156 = 0.019 TJ

The value of $BC_{p,j,k,y}$ is derived from the project KPT and $Hh_{j,k}$ (same as Hh_i) is derived from the Baseline KPT in line with the applied methodology.

BE_y = Min ($EC_{i,y}$, $EC_{est,y}$) × $N_{j,k,y} \times n_{j,k,y} \times (EF_{b,i,CO2} \times f_{NRB,y} + EF_{b,i,nonCO2})$

= 0.62 × 335,314 × 0.9 × (112 × 0.8872 × (1-26%) + 9.46)

= 1,544,826 tCO_2e (Estimated Baseline Emissions for the year 2024)

Where, $N_{j,k,y}$ = 335,314 (Actual distribution till the end of year 2024)

$n_{j,k,y}$ = 0.9 (assumed for ex-ante estimation only)

EF_{CO2} = 112 tCO2e/TJ (Meth Default Value)

$EF_{non\text{-}CO2}$ = 9.46 tCO2e/TJ (Meth Default Value)

f_{NRB} = f_{NRB} (as per registered PD, refer appendix 2, tool 30 v 4.0) × (1-Adjustment factor of 26%) = 0.6565

4.2 Project Emissions

As per the applied methodology VM0050, ver. 1.0, project emissions are calculated as follows:

$$PE_y = PE_{energy,y} + PE_{others,y} \quad \text{Equation (6)}$$

Where:

PE_y	Project emissions during year y (t CO_2e)
$PE_{energy,y}$	Project emissions from energy consumption of project devices in year y (t CO_2e)
$PE_{others,y}$	Project emissions from other sources in year y (t CO_2e)

$PE_{energy,y} = \sum_j \sum_k BCp,j,k,y \times Nj,k,y \times NCVp,j \times nj,k,y \times (EFp,j,CO2 \times fNRB,y + EFp,j,nonCO2)$ Equation (7)

Where:

$EF_{p,j,CO2}$	CO_2 emission factor for fuel used by project device type j in the project scenario (t CO_2/TJ) = 112 tCO_2e/TJ (Meth Default Value)
$EF_{p,j,nonCO2}$	Non-CO_2 emission factor for fuel used by project device type j in the project scenario (t CO_2e/TJ) = 9.46 tCO_2e/TJ (Meth Default Value)

Given that the grouped project involves the distribution of non-renewable biomass-based improved cookstoves exclusively to the target users, Equation (8) and Equation (10) of the methodology applies to electric stoves and are therefore not applicable. Further, the ICS distributed under the project is designed for wood fuel, and beneficiaries procure the wood fuel either from nearby fields or the market. Therefore, Equation (9) is not applicable.

<u>Sample Calculations of Estimated Project Emissions for the Year 2024 of CP1:</u>

Parameter	Applied value/Approach
$PE_{energy,y}$	= $BC_{p,j,k,y}$ (tons/person/year) × $Hh_{j,k}$ × $N_{j,k,y}$ × $NCV_{p,j}$ × $n_{j,k,y}$ × ($EF_{p,j,CO2}$ × $f_{NRB,y}$ + $EF_{p,j,nonCO2}$) = 0.282 × 4.31 × 335,314 × 0.0156 × 0.9 × (112 × 0.6565 + 9.46) = 475,372 tCO_2e The value of $BC_{p,j,k,y}$ is derived from the project KPT and $Hh_{j,k}$ (same as Hh_i) is derived from the Baseline KPT in line with the applied methodology.
$PE_{others,y}$	= 0
PE_y	= $PE_{energy,y}$ + $PE_{others,y}$ = 475,372 tCO_2e (Estimated Project Emissions for the year 2024)

4.3 Leakage Emissions

Leakage shall be considered as default 0.95 in accordance with the methodology.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

As per the applied methodology VM0050, ver. 1.0, Net GHG emission reductions are calculated as follows:

$$ER_y = (BE_y - PE_y) \times 0.95 - LE_{RB,y} \qquad \text{Equation (11)}$$

Where:

ER_y	Emission reductions during year y (t CO_2e)
$LE_{RB,y}$	Leakage emissions associated with use of renewable biomass during year y (t CO_2e)

Given that the grouped project involves the distribution of non-renewable biomass-based improved cookstoves exclusively to the target users, leakage emissions associated with the use of renewable biomass are not applicable, as the project does not involve renewable biomass as a fuel source.

Sample Calculations of Estimated Net GHG Emission Reductions for the Year 2024 of CP1:

ER_y $= (BE_y - PE_y) \times 0.95 - LE_{RB,y}$

= (1,544,826 - 475,372) × 0.95 - 0

= 1,015,981 tCO_2e (Estimated Net GHG Emission Reductions for the year 2024)[23]

For detailed calculations on the estimated VCUs throughout the crediting period refer Ex-ante ER Sheet.

Table: Estimated Emission Reductions for the first crediting period

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
04-Nov-2022 to 31-Dec-2022	98,884	29,733	-	65,693	-	65,693
01-Jan-2023 to 31-Dec-2023	1,574,347	474,074	-	1,045,259	-	1,045,259
01-Jan-2024 to 31-Dec-2024	1,544,826	475,372	-	1,015,981	-	1,015,981
01-Jan-2025 to 31-Dec-2025	2,014,618	609,301	-	1,335,051	-	1,335,051
01-Jan-2026 to 31-Dec-2026	2,176,264	672,492	-	1,428,583	-	1,428,583
01-Jan-2027 to	2,172,511	685,914	-	1,412,267	-	1,412,267

[23] The sample calculations are for Kuniokoa ICS distribution, and ERs are calculated for year 2024 of operations.

31-Dec-2027						
01-Jan-2028 to 31-Dec-2028	2,110,856	680,915		1,358,443	-	1,358,443
01-Jan-2029 to 03-Nov-2029	1,715,651	565,439		1,092,701	-	1,092,701
Total	**13,407,957**	**4,193,240**	-	**8,753,978**	-	**8,753,978**

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$EF_{b,i,CO2}$ $EF_{p,j,CO2}$
Data unit	t CO_2/TJ
Description	CO2 emission factor for fuel used by baseline device type i in the baseline scenario CO2 emission factor for fuel used by project device type j in the project scenario
Source of data	Option 3: Default value from the most recent version of the *IPCC* Guidelines for National Greenhouse Gas Inventories
Value applied	Wood: 112 t CO_2/TJ
Justification of choice of data or description of measurement methods and procedures applied	Use of default values from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories Wood: 112 t CO_2/TJ
Purpose of data	Calculation of baseline and project emissions
Comments	N/A

Data / Parameter	$EF_{b,i,non\text{-}CO2}$ $EF_{p,j,non\text{-}CO2}$

Data unit	t CO_2e/TJ
Description	Non-CO2 emission factor for fuel used by baseline device type i in the baseline scenario Non-CO2 emission factor for fuel used by project device type j in the project scenario
Source of data	Option 3: Default value from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	Wood: 9.46 t CO2/TJ
Justification of choice of data or description of measurement methods and procedures applied	Use of default values from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories Wood: 9.46 t CO2e/TJ (AR5 GWP)
Purpose of data	Calculation of baseline and project emissions
Comments	N/A

Data / Parameter	$NCV_{b,i}$ $NCV_{p,j}$
Data unit	TJ/tonne
Description	Net calorific value of baseline fuel used by baseline device type i Net calorific value of project fuel used by project device type j
Source of data	Option 3: Default value from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories
Value applied	Wood: 0.0156 TJ/tonne
Justification of choice of data or description of measurement methods and procedures applied	Use of default values from the most recent version of the IPCC Guidelines for National Greenhouse Gas Inventories Wood: 0.0156 TJ/tonne
Purpose of data	Calculation of baseline and project emissions
Comments	N/A

Data / Parameter	$BC_{ex\text{-}ante,b,i}$[24]
Data unit	Tonnes/year

[24] This parameter corresponds to BCex-ante,b,i for the first two years and where the follow-up baseline survey campaign shows that there are no significant changes in baseline fuel consumption. Otherwise, it corresponds to $BC_{b,i,y}$.

Description	Ex-ante annual average quantity of fuel used per baseline device type *i*
Source of data	**Option 1:** A measurement campaign (Baseline KPTs), conducted following the Kitchen Performance Test Protocol and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities.
Value applied	4.04
Justification of choice of data or description of measurement methods and procedures applied	This parameter has been fixed ex-ante. A measurement campaign (Baseline KPTs), conducted following the Kitchen Performance Test Protocol and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities. The survey revealed that the average wood consumption in the Rwanda is 0.937 tonnes/person/year. This result is scaled appropriately using the average household size (Hh_i) as follows: $BC_{ex\text{-}ante,b,i}$ = $BC_{ex\text{-}ante,b,I}$ (tonnes/person/year) x Equivalent standard male adults (Hh_i) obtained from baseline KPT = 0.937 x 4.31 = 4.04 tonnes/year The details of the baseline KPTs including target population, sample size, sampling method, precision/confidence achieved has been added in section 3.4.
Purpose of data	Calculation of baseline emissions
Comments	This parameter has been fixed ex-ante for the next 5 years. -

Data / Parameter	$\eta_{old,avg}$
Data unit	Fraction
Description	Weighted average efficiency of baseline devices that are replaced by project devices
Source of data	VM0050 Default value
Value applied	0.15
Justification of choice of data or description of measurement methods and procedures applied	This data is fixed ex-ante as per the applied methodology VM0050. A default value of 0.15 shall be used if the baseline device is a three-stone fire using firewood or a cookstove without improved combustion air supply or flue gas ventilation. The project distributes non-renewable wood fuel-based ICS, replacing only baseline devices that meet these criteria.

Purpose of data	Calculation of baseline emissions
Comments	-

Data / Parameter	Hh_i $Hh_{j,k}$
Data unit	Equivalent standard male adults
Description	Average household size of the target population using device type i Average household size of the target population using device type j from batch k
Source of data	Baseline survey /KPT
Value applied	4.31
Justification of choice of data or description of measurement methods and procedures applied	A measurement campaign (Baseline KPTs/Surveys), conducted following the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities.
Purpose of data	Estimation of average energy consumption when applying Option 1: Measurement campaign (Section 8.1.1). Cross-checking energy and fuel consumption values.
Comments	N/A

5.2 Data and Parameters Monitored

Data / Parameter	$N_{j,k,y}$
Data unit	Number
Description	Number of commissioned project devices of type *j* from batch *k* in year *y*
Source of data	Monitoring
Description of measurement methods and procedures to be applied	The following data must be recorded during project activity implementation: 1) Number of new devices distributed under the project activity, identified by the type of device and date of commissioning; and 2) Identification information of the recipient of the device distributed under the project activity (e.g., name, address, phone number). Data management and reporting of this information must adhere to both data privacy requirements and good practice.

Frequency of monitoring/recording	Every time that new project devices are distributed
Value applied	500,000 (for ex-ante estimation only)
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of baseline and project emissions
Calculation method	N/A
Comments	The number of project devices will be recorded in a database, sales record, or similar to ensure transparency. Here, the batch k is not segregated in a PAI. Every distribution under a PAI may be considered as a separate batch.

Data / Parameter	$n_{j,k,y}$
Data unit	Fraction
Description	Proportion of commissioned project devices of type *j* from batch *k* that are still being used regularly in year *y*
Source of data	Monitoring
Description of measurement methods and procedures to be applied	**Option 1 (SUMs):** Measured directly using stove use monitors (SUMs) in a sample of users according to the most recent version of the CDM Standard for Sampling and Surveys for Project Activities and Programmes of Activities. **Option 2 (surveys):** Based on an adoption rate determined by a survey according to the most recent version of the CDM Standard for Sampling and Surveys for Project Activities and Programmes of Activities.
Frequency of monitoring/recording	**Option 1: (SUMs):** continuous **Option 2 (surveys):** annually
Value applied	0.9 (for ex-ante estimation only)
Monitoring equipment	N/A
QA/QC procedures to be applied	The date on which a sample project device stopped being used should be taken as follows: Option 1: (SUMs): The date on which the SUM ceased registering any activity of the project device Option 2 (surveys): Where the project device is not working or not being used at the time of conducting the survey, it should be conservatively assumed that the project device has not been

	active since the date on which the last adoption survey was conducted.
Purpose of data	Calculation of baseline and project emissions
Calculation method	N/A
Comments	The batch k is not segregated in a PAI. Every distribution under a PAI may be considered as a separate batch.

Data / Parameter	$f_{NRB,y}$
Data unit	Fraction or %
Description	Fraction of woody biomass that is established to be non-renewable used by device in year y
Source of data	Third party report in line with TOOL30 version 4.0 and applying an uncertainty discount of 26%
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	Ex-ante for each crediting period
Value applied	0.6565
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of baseline and project emissions
Calculation method	The value of 0.8872 was registered in the PD under TOOL30 Version 4 for the host country, Rwanda. This registered value in the PD is then adjusted by an uncertainty discount of 26% in line with VM0050 as follows: = 0.8872 x (1-26%) = 0.6565
Comments	This data has been fixed for entire crediting period. The applied f_{NRB} value has been calculated by using the latest available data sources. As per the fNRB calculation done by PP, the value is 91% however as a measure of conservativeness, PP used the fNRB value from another registered projects which was lesser than 91%. Hence, the value has been changed to 88.72% to which the uncertainty discount has subsequently been applied.

Data / Parameter	$BC_{b,i,y}$
Data unit	Tonnes/year
Description	Fuel used per baseline device type *i* during year *y*

Source of data	**Option 1:** A measurement campaign following the Kitchen Performance Test Protocol in control households that do not participate in the project, established prior to validation as statistically equivalent to the pre-project conditions of project households regarding baseline fuel consumption
Description of measurement methods and procedures to be applied	**Option 1:** Follow-up baseline surveys must be conducted at most every five years from the date of the last survey in control households that do not participate in the project, established prior to validation as statistically equivalent to the pre-project conditions of project households regarding baseline fuel consumption. Measurement campaign must be updated where follow-up baseline surveys show that the fuels, fuel sources, or technologies used by the control group are no longer statistically equivalent to the pre-project conditions of project households. The measurement campaign must be designed, carried out, and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities. The result must be scaled appropriately using the average household size to obtain the value of $BC_{b,i,y}$.
Frequency of monitoring/recording	At most every five years
Value applied	4.04
Monitoring equipment	N/A
QA/QC procedures to be applied	The campaign must achieve a confidence and precision of at least 90/10 for the target parameter of average daily fuel consumption per adult equivalent. Where the project does not achieve the target precision in a monitoring period, the project proponent must apply an appropriate conservativeness deduction as per Section 4 of the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities.
Purpose of data	Calculation of baseline emissions
Calculation method	N/A
Comments	The PP has allocated 128 baseline surveyed households as control households. On baseline reassessment in 5years time the PP will contact these same Households for re-assessment. If the household has moved out of the geographic area, or if the household is not available for any reason the PP will add in new households to the control group from within the same village,

	utilising again the random walk method. This will ensure sample size integrity and continued reflection of baseline conditions within the project area.

Data / Parameter	$\eta_{new,avg,y}$
Data unit	Fraction
Description	Weighted average efficiency of project devices in year *y*
Source of data	Monitoring
Description of measurement methods and procedures to be applied	The efficiency must be established using one of the following methods, and the corresponding documentation must be presented: 1) Water Boiling Test campaigns achieving 90/10 confidence and precision levels as per the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities; 2) Manufacturer-certified value that is determined via Water Boiling Test; or 3) Certification from the host country's national standards body or certifying agency. The decrease in thermal efficiency of project device j from batch k due to aging must be accounted for during the monitoring period, as presented below. For devices using biomass or fossil fuel, one of the following options must be selected, listed in descending order of preference: a) Standard Water Boiling Test campaigns[25] b) A linear decrease approach, applying a default schedule of linearly decreasing efficiency up to the terminal efficiency (assumed to be 25%) through the lifespan of the project device. The weighted average efficiency is calculated by multiplying the energy consumption $EC_{p,j,k,y}$ of the project device j of batch k by its thermal efficiency $\eta_{new,j,k,y}$ and then summing these values for all project devices. The resulting sum is divided by the total energy consumption of all project devices to determine the weighted average efficiency.
Frequency of monitoring/recording	Annually

[25] Must be carried out following national standards (where available) or international standards or guidelines.

Value applied	The following are the efficiency values of the example ICS based on certification from the certifying agency or manufacturer specifications: <table><tr><td>ICS Model</td><td>Efficiency with Pot-skirt</td><td>Efficiency without Pot-skirt</td></tr><tr><td>Kuniokoa</td><td>51.30%</td><td>43.40%</td></tr><tr><td>Ecoa Wood</td><td>52.00%</td><td>45.50%</td></tr></table>An efficiency loss of 2% is assumed for ex-ante estimation, while the actual value will be monitored ex-post annually.
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of baseline and project emissions
Calculation method	N/A
Comments	Different models of ICS may be used at the time of distribution as indicated in Section 1.12 above. In case of inclusion of other/new models, during the project lifetime, corresponding thermal efficiency values shall be used for Ex-post ER calculations.

Data / Parameter	$BC_{p,j,k,y}$
Data unit	Tonnes/year
Description	Average quantity of fuel used by project device type j from batch k during year y
Source of data	Monitoring
Description of measurement methods and procedures to be applied	**Option 1: Kitchen Performance Test (KPT)** A measurement campaign following the Kitchen Performance Test Protocol protocol must be designed, carried out, and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities. The campaign must achieve confidence and precision of at least 90/10 for the target parameter of average daily fuel consumption per adult equivalent. The result must be scaled appropriately using the average household size to obtain the value of $BC_{p,j,k,y}$. Project activities applying KPTs must also measure the quantity of fuel used by pre-project devices that are still operational and provide it in the calculation sheet and monitoring reports (this is a reporting-only requirement). **Option 2: Direct measurement**

	Apply continuous direct measurement using equipment calibrated in accordance with national/international requirements. A sample of project devices may be measured in such a way that confidence and precision of 90/10 is achieved for the target parameter of total annual fuel use. The sampling must comply with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities.
Frequency of monitoring/recording	Biennial or annual for Option 1 Continuous and aggregated annually for Options 2 and 3
Value applied	1.21
Monitoring equipment	N/A
QA/QC procedures to be applied	Where SUMs are used to measure project stove adoption, the stove usage indicated by the measurements for this parameter must be consistent with the frequency of use indicated by SUM measurements.
Purpose of data	Calculation of baseline and project emissions
Calculation method	A measurement campaign (Project KPTs), conducted following the Kitchen Performance Test Protocol protocol and analyzed in compliance with the most recent version of the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities. The survey revealed that the average wood consumption on the project ICS is 0.282 tonnes/person/year. This result is scaled appropriately using the average household size ($Hh_{j,k}$) as follows: $BC_{p,j,k,y}$ = $BC_{p,j,k,y}$ (tonnes/person/year) x Equivalent standard male adults ($Hh_{j,k}$) obtained from project KPT = 0.282 x 4.31 = 1.21 tonnes/year
Comments	N/A

5.3 Monitoring Plan

This Grouped Project in Rwanda follows VCS standard requirements applicable, along with the methodological requirements on monitoring a grouped non-AFOLU large project activity. The Monitoring Plan is elaborated in multiple sub-sections for the sake of clarity. This monitoring plan is based on the applied methodology VM0050 ver 1.0. The monitoring plan shall consist of checking a representative sample of all ICS units distributed to ensure that they are still operating or are replaced by an equivalent in-service unit.

1. **Methods for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters**

A project database will be populated during the distribution of each initial cookstove issued, and will be updated based on subsequent replacements, as well as detailed data on the representative sample surveyed for monitoring purposes. This database for the project activity will be maintained by the DelAgua. The database will be accessible to the project proponent, appropriate partners, and the verification VVB. The database will include, at minimum, the following:

- Date of Distribution
- Geographic area of distribution;
- Model/type of project technology distributed
- Name, telephone number (where available), and address of recipient; and
- Unique alpha/numeric ID for each device that is distributed

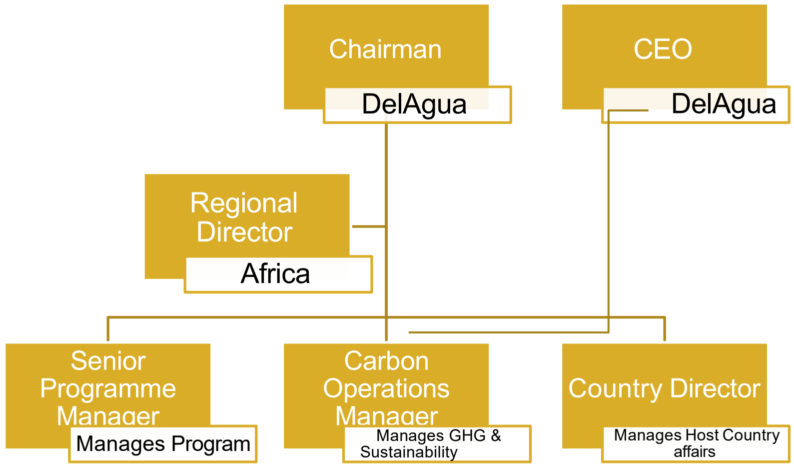
The database will be available to select a random, representative sample for monitoring and verification purposes. This sample set will be integrated into the database to include additional monitoring parameters as required or as appropriate. The data will be collected using surveys carried out either by external third parties or by project proponent own teams. Monitoring surveys will include examination of the cookstove installations and verification that they are in an operational condition. Data collected from the surveys will be compiled into an Excel spreadsheet. Data obtained from the samples will be used to estimate proportions and mean values for the parameters described in below sub-sections. The values will then be factored into the emissions reduction calculations.

The data that will be recorded in the monitoring surveys is collected and stored electronically, and then uploaded to the central database management system, where it is maintained online on a subscription-based server host.

The data will be also stored offline, within the hard disk drives. The data collection procedures do not involve any paper records, to reduce the environmental footprint of the programme.

The project related data will be stored for two years after the implementation period is over for the project activity.

2. Organizational Structure, Role, and Responsibilities of Personnel Involved in the Project Activity



The DelAgua team involved in the GHG operations management is the local staff based in Rwanda along with the senior management.

The key responsibilities are shared by the team as follows:

Chairman – Supervises the overall programme at strategy and financial side.

CEO – Acts as central point of strategy, programme management, finance, and technical domains of the programme.

Regional Director – Responsible for the regional level programme implementation and expansion planning and implementation.

Senior Programme Manager – Manages the programme at implementation level, and acts as an internal technical reviewer for the programme documents and procedures.

Carbon Operations Manager – Manages the operations for GHG emission mitigation programme, from planning to implementation stage. Additionally manages the Sustainability aspects of the programme.

Country Director – Supports programme in the operations and management at the host country level and acting as point of communication between DelAgua and government of Rwanda.

The collective responsibilities of DelAgua as the Project Participant is as described below:

- Acts as the project managing entity
- Development of PD
- Registration/Issuances of the project with VERRA
- Inclusion of new PAIs upon satisfaction of the eligibility criteria stipulated in the PD
- Authorized entity for any official communication with the VERRA and VVB

- Manage and archive the database for all appliances
- QA/QC of database during distribution & data collection
- Maintain local repair/replacement facilities
- Education and public awareness raising during initial distribution as well as periodic education campaigns.
- Data collection during distribution & delivery of data to database
- QA/QC of data collection
- Conduct ongoing monitoring, as required to ascertain monitored parameters.

3. **The procedures for Internal Auditing and QA/QC**

Quality assurance and quality control (QA/QC) measures will be adhered as described in section 5.2. Distributors are trained in the proper installation, use, and maintenance of the devices, and are provided with training sufficient to introduce recipient households to the proper operation and maintenance of the devices, as well as contact information for DelAgua. Supervisors receive similar training, as well as supervisory training in monitoring of distributor's activities. Supervisors are responsible for approximately 10 distributors. During the campaign, DelAgua staff will spot-check distributor quality, by revisiting 2 households per distributor, to verify that the stoves and /or lamps are in place, and that the recipient can show proper operation of the units. The results of this spot check are recorded electronically and compared against the initial distribution logs.

The data collected in the database is verified and controlled in the following way:

1. Surveys are conducted by staff trained by DelAgua.
2. Supervisors are separately trained by DelAgua to spot-check surveyor results.
3. All ex-ante and monitored data parameters are collected on smart-phone based surveys and are uniform across all surveyors for any given survey activity.
4. Unknown to the surveyors, the database analysis includes verification of "valid" surveys including by a.) verifying household consent is received and b.) that the survey duration, as recorded by Internet time, is no shorter or longer than a reasonable value. For example, the baseline surveys for Rwanda were validated against a minimum of 20 minutes and a maximum of 120 minutes.
5. Data validation is conducted on the database to ensure that values are numerically consistent, i.e. multiple-choice values add up to 100% of respondents.

4. **Procedures for handling non-conformances with Validated Monitoring Plan**

The personnel engaged in monitoring and reporting will undergo training to ensure a comprehensive understanding and proper execution of the monitoring and reporting procedures outlined in the monitoring plan. DelAgua is responsible for training any contractors used during distribution, education and monitoring activities. DelAgua ensures training of all on-site staff with respect to adherence to the Monitoring Plan of the project activity. Records of the training will be kept. Training records are saved for such training programs, where step-by-step training is provided to the surveyors for a quality data collection process, where they are expected to not impact the materially, degree of conservativeness, and perform the surveys in an unbiased manner. The surveyors are trained on a periodic basis, and refresher trainings are also provided to improve the quality of surveys from time to time. Since the surveys are electronic in format, recording of responses and the record keeping procedure for these surveys are conveyed to the team through rigorous trainings.

The project proponent ensure that field personnel have reviewed, understood and have agreed to follow the monitoring plan procedures, including provisions for maximizing response rates, documenting out-of-population cases, refusals and other sources of non-response. Monitoring plan explains how to properly survey households to prevent bias from interviewer. In the case a household refuses to participate, another household is chosen at random. To reduce interviewer bias, good questionnaire design and well-tested questionnaires is used.

5. Sampling Plan

Due to the large number of ICS envisioned to be distributed as part of the project activity, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling will be undertaken as part of Sampling Plan in line with the requirements of the methodology applied and the Standard: Sampling and surveys for CDM project activities and programmes of activities, version 9.0

The sampling plan is intended to enable the discovery of unbiased and reliable estimates of monitored parameter values used in the GHG emission reduction calculations. The sampling plan design is based on the CDM Guideline: Sampling and surveys for CDM project activities and programmes of activities, version 4.

Sample sizes will be sufficient to ensure that the precision of the sample means/proportions are in accordance with the methodological / standard requirements. In cases where survey results indicate that desired precision is not achieved, the lower bound value of the corresponding confidence interval of the parameter value may be used as an alternative to repeating the survey. Alternatively, the survey may be expanded to reach the required confidence/precision.

The sampling methodology will be in accordance with the representative sampling methods provided by the sampling guidelines and standards as indicated along this section, with the applied methodology having precedence.

STEP 1: Sampling Objectives

The sampling objective for each parameter is to determine, via sampling survey, a statistically significant parameter value for the emission reduction calculations. These parameters are as listed in section 5.2:

No.	Parameter	Description	Parameter type	Monitoring frequency
1	$n_{j,k,y}$	Proportion of commissioned project devices of type j from batch k that remain operating in year y	Proportion	Annually
2	$\eta_{new,avg,y}$	Weighted average efficiency of project devices in year y	Mean	Annually
3	$BC_{b,i,y}$	Fuel used per baseline device type i during year y	Mean	Every five years
4	$BC_{p,j,k,y}$	Average quantity of fuel used by project device type j from batch k during year y	Mean	Biennial or annual

STEP 2: Target Population

The target population refers to the total number of project devices commissioned under the project.

STEP 3: Sampling Frame

The sampling frame refers to all the aggregated data of end-users of the ICS as recorded in the Project ICS Database.

STEP 4: Sample Method

Sampling will be conducted using stratified random sampling techniques[26] over the sampling frame, and detailed calculations are provided below as per CDM guidelines “Sampling and surveys for CDM project activities and programmes of activities”. The ICSs in the sampling frame shall be stratified as per parameter requirements. Stratified Random Sampling will be used to select samples from the Project Database for monitoring parameters.

Optionally, other sampling approaches may be used in accordance with Standard “Sampling and surveys for CDM project activities and programmes of activities” and Guideline for Sampling and Surveys for CDM Project Activities and Programme of Activities when sampling techniques or statistical analysis necessitates it.

To ensure a random selection of ICS, random number generators shall be applied. Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame. Applying the

[26] Other Sampling approaches may be applied during the course of crediting period, in case needed.

random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size as calculated by the project proponent.

STEP 5: Desired Precision / Expected Variance and Sample Size

The sample size shall be determined using the following formula:

$$n \geq \frac{z^2 * N * V}{(N-1) * precision^2 + z^2 * V}$$

Where:

Parameter	Description
n	Sample Size
N	Total number of cookstoves (ICS) distributed
Z	Constant referring to level of confidence (1.96 for 95 %, 1.646 for 90%)
Precision	Required precision (e.g., 10% = 0.1)

Where for Proportion Parameter:

$$V = \frac{SD^2}{\overline{p}^2}$$

$$SD^2 = \frac{\sum_{i=1}^{k} g_i * p_i * (1 - p_i)}{N}$$

$$\overline{p} = \frac{\sum_{i=1}^{k} g_i * p_i}{N}$$

Parameter	Description
SD	Standard deviation
g_i	Weight of strata i in the population
p_i	Expected proportion of strata i in the population
k	Total number of strata in the population

And for mean-based parameters:

$$V = \frac{SD^2}{mean^2}$$

$$SD^2 = \frac{\sum g_i \times SD_i^2}{N}$$

$$mean = \frac{\sum g_i \times m_i}{N}$$

Parameter	Description

Mean	Weighted overall mean
m_i	Mean of strata i

To calculate estimates of the parameters of interest (proportion, mean, and standard deviation) required for sample size calculations, different approaches may be used in line with the Guidelines for Sampling and Surveys for CDM Project Activities and Programmes of Activities, Version 04.0, Appendix 1, Paragraph 5, as follows:

(a) Refer to the results of previous studies and use these findings.

(b) If information from previous studies is unavailable, conduct a preliminary sample as a pilot study and use that sample to derive estimates.

(c) Use best estimates based on the researcher's own experience.

Step 6: Quality Assurance/Quality Control

During sampling, there may be non-response from the target population. Over-sampling may be used to avoid non-response as a part of QA/QC; however, sampling may be ceased once required confidence/precision is met.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

No information presented in the project description is deemed commercially sensitive.

APPENDIX 2: FNRB CALCULATION

Step 1: Estimation of renewable biomass (RB) in the country of Rwanda as per equation (4) of the fNRB tool

$$RB = \sum \left(MAI_{forest,i} \times (F_{forest,i} - P_{forest,i})\right) + \sum (MAI_{other,i} \times (F_{other,i} - P_{other,i})) \quad \text{Equation (4)}$$

Where:

$MAI_{forest,i}$	=	Mean Annual Increment of woody biomass growth per hectare in sub-category *i* of forest areas in the relevant period (tonnes/ha/yr)
$MAI_{other,i}$	=	Mean Annual Increment of woody biomass growth per hectare in sub-category *i* of other land areas in the relevant period (tonnes/ha/yr)
$F_{forest,i}$	=	Extent of forest in sub-category *i* in the relevant period (ha)
$F_{other,i}$	=	Extent of other land in sub-category *i* in the relevant period (ha)
$P_{forest,i}$	=	Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within forest areas (in sub-category *i*) in the relevant period (ha)
$P_{other,i}$	=	Extent of non-accessible area (e.g. protected area where extraction of wood is prohibited, geographically remote area) within other land areas (in sub-category *i*) in the relevant period (ha)
i	=	Sub-category *i* of forest areas and other land areas[3]

Parameters used:

Input parameter	Value	Data source
Extent of forest and other wooded land in 2021 (in ha), $F_{forest,i}$	**467,239**	Geospatial data products for Rwanda were analysed to estimate Rwanda's renewable biomass. The forest and other wooded land cover for 2000 and 2021 was estimated using Hansen/UMD/Google/USGS/NASA[27] spatial data for 2000 and 2021, which is derived from Hansen et al.[28], and disaggregated according to the FAO[29] global ecological zones. The total woody cover extent, including forests and other wooded areas, was calculated for each ecological zone, within the protected areas and within areas that are either accessible or geographically remote.
Protected areas within forest areas (in ha)	137,289	
Remote areas within forest areas (in ha)	12,681	

27 Hansen/UMD/Google/USGS/NASA. Hansen Global Forest Change v1.9 (2000-2021). Available at: https://glad.earthengine.app/view/global-forest-change. Accessed on: 31 May 2023.

28 Hansen, M. C. C. et al. High-resolution global maps of 21st-century forest cover change. Science (80- .). 342, 850–854 (2013)

29 FAO. 2012. Global ecological zones for FAO forest reporting: 2010 Update. Forest Resources Working Paper 179. Food and Agriculture Organisation of the United Nations, Rome. Available at: https://www.fao.org/3/ap861e/ap861e00.pdf.

Extend of non-accessible area (in ha), $P_{forest,i}$	**149,970**	Calculated (protected forest + remote forest)
Mean Annual Increment (t/ha/year), $MAI_{forest,i}$	1.82	The mean annual increment for forest is calculated as default age weighted average of the forest type and as per IPCC[30] values for Tropical Mountain system.

Geographically remote areas were determined to be areas beyond a harvestable distance from roads and settlements. Conservatively, it was assumed that forests and wooded areas adjacent to all roads within a distance of 2.5 km are harvestable, despite how geographically remote or inaccessible these roads may be. This threshold was determined based on the peer-reviewed literature on wood harvesting practices.

Banks et al.18 found that woody biomass density increases significantly as a function of distance from the edge of a settled area. Beyond 450 m, average biomass density was found to increase from 2.1 to 16.6 t/ha. Jumbe and Angelsen[31] modelled fuelwood choice from household survey data. They found that the most significant determinants of fuelwood choice are source attributes (size and species composition) and distance to source. The mean harvesting distance for their study was found to be 0.95 km. Bandyopadhyay et al.[32] matched household survey and remote sensing data to investigate wood fuel use in relation to poverty levels. They concluded that the one-way walk duration to wood fuel source for both poor and non-poor households is 0.6 hours. Assuming an average walking speed of 4 km/hr over uneven terrain, the average one-way distance to wood fuel source is less than 2.5 km. Areas beyond the harvestable distance were, therefore, determined to be geographically remote. Accessible areas are those that are not protected and not geographically remote, from which the annual renewable biomass production is calculated.

The default age-weighted mean annual increment (MAI) estimates of each ecological zone, as reported by the IPCC, were used for this study. The proportion of forest stand ages above and below 20 years old, including primary forests, were estimated for each ecological zone by extrapolating the observed forest gain extents between 2000 and 2012 to a 21-year period. The average MAI estimates for Rwanda is 1.82 t/ha/year for tropical mountain system between 2000 and 2012 to a 20 years period.

FAO Global Ecological Zone	**Total forest cover** (ha)	**Protected area cover** (ha)	**Remote area cover** (ha)	**MAI** (t/ha/yr)	**Annual growth, RB**(t/yr)
Tropical mountain system	467,239	137,289	12,681	1.82	578,495
Total	**467,239**	**137,289**	**12,681**	-	**578,495**

Total renewable biomass RB = 578,495 t/year

[30] IPCC. Forest Land. 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4: Agriculture, Forestry and Other Land Use 4, 1–29 (2019)

[31] Jumbe, C. B. L. & Angelsen, A. Modeling choice of fuelwood source among rural households in Malawi: A multinomial probit analysis. Energy Econ. 33, 732–738 (2011).

[32] Bandyopadhyay, S., Shyamsundar, P. & Baccini, A. Forests, biomass use and poverty in Malawi. Ecol. Econ. 70, 2461–2471 (2011)

Step 2: Estimation of consumption of woody biomass (H) in the country of Rwanda as per equation (2) of the fNRB tool

NRB = H - RB

Where,

H = Total consumption of woody biomass in the applicable area in the relevant period (tonnes)

H = HW x N + CE + NE

Where,

HW = Average consumption of wood fuel per household, including fuelwood and charcoal, in the applicable area in the relevant period (tonnes/household)

N= Number of households consuming wood fuel within the applicable area in the relevant period (number)

CE= Commercial woody biomass consumption for energy applications (e.g. commercial, industrial or institutional uses of woody biomass in ovens, boilers etc.) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

NE= Commercial woody biomass consumption for non-energy applications (e.g. construction, furniture) that are extracted from forests or other land areas in the applicable area in the relevant period (tonnes)

The total woody biomass consumption combines the country's domestic and non-domestic *energy* and non-domestic *non-energy* consumption. Rwanda's estimate of total woody biomass consumption estimation is derived from World Bank population statistics[33] and consumption data sourced from literature[34,35], the FAO[36] and KPT surveys. Reported values of charcoal consumption are converted to the equivalent woody biomass using guidelines on kiln efficiency from the FAO forestry paper on industrial charcoal making[37] and the latest CDM Tool 30 v04.0 2022 default charcoal-to-wood biomass conversion factor. The CDM default value of 4 was applied to the urban domestic charcoal consumption and non-domestic energy charcoal consumption.

Data was drawn from the most recently available official reports/statistics, in this case from the UN's Statistics Division to demonstrate both the recent trend in increase of wood fuel use at the household level, plus also the most recently available data (from 2021). This increase in fuelwood use at the household level has had a significant impact on the parameter value fNRB. The density of fuel is not a time bound parameter, hence average of Africa is used as per published data by FAO.

Input parameter	Value	Data source

33 https://data.worldbank.org/.
34 United Nations & African Union. Review of woodfuel biomass production and utilization in Africa
35 FAO. Forestry Sector in 2020: https://www.fao.org/forest-resources- assessment/fra-2020/country-reports/en/.
36 FAO Statistics. Forest Products 2019. https://www.fao.org/3/cb3795m/cb3795m.pdf (2021)
37 Charcoal transition: https://www.fao.org/3/i6935e/i6935e.pdf

Wood fuel consumption (m^3/capita/yr)	0.396	FAO[38]
Population of the country	13,276,517	World Bank[39]
Wood fuel consumption (m^3/yr)	**5,257,220**	**Calculated**
Conversion factor (t/m^3)	0.725	FAO[40]
HW (Average consumption of wood fuel tonnes/yr)	**3,811,384**	**Calculated**

Input parameter	**Value**	**Data source**
H (in t/year)	6,177,233	Calculated
HW woody biomass consumption in households (in t/year)	3,811,484	Calculated
CE Non domestic woody biomass consumption for energy application (in t/year)	201,877	Calculated, using wood to charcoal conversion factor[41]
NE Non domestic woody biomass consumption for non-energy application (in t/year)	2,163,872	Calculated using biomass expansion factor 1.50
Population of the country	13,276,517	World Bank[42]

Equation (3) of the fNRB tool stipulates that the average household wood fuel consumption is multiplied with the number of households and industries consuming wood fuel for energy and non-energy applications within the country. Since most of the available and reliable data is total wood fuel consumption in households and commercial and other use.

The total population of host country has been sourced as per the published World Bank population data from 2022. This is in line with the fNRB tool.

Step 3: Estimation of quantity of non-renewable biomass (t/year) as per equation (2) of the fNRB tool

NRB = H - RB

Where,

[38] https://www.fao.org/faostat/en/#data/FO
[39] https://data.worldbank.org/
[40] http://www.fao.org/3/a-i6935e.pdf
[41] https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-33-v2.0.pdf

[42] https://data.worldbank.org/

H = Total consumption of woody biomass in the applicable area in the relevant period (tonnes)

This step calculates the quantity of non-renewable biomass (t/year) (NRB) by subtracting the renewable biomass estimated in step 1 from the total consumption of woody biomass estimated in step 2. No further input parameters are needed under this step.

NRB = 6,177,233– 578,495

NRB = 5,598,738 t/year

Step 4: Estimation of fraction of woody biomass that can be established as non-renewable (fNRB) as per equation (1) of the fNRB tool

$$fNRB = \frac{NRB}{NRB + RB}$$ Equation (1)

This step calculates the fraction of woody biomass that can be established as non-renewable (fNRB). The non-renewable biomass calculated in step 3 is divided by the sum of non-renewable biomass and renewable biomass calculated in step 1. No further input parameters are needed under this step.

fNRB = 578,495/ (5,598,738 + 578,495)

fNRB = **91.0%**

Step 5: Comparison as per para 6 of CDM Tool 30 version 4

As per the Bailis report, the f_{NRB} value for Rwanda is 59%. Rwanda is considered as wood fuel "hotspot" as most of the harvested wood fuel is unsustainable. The resulting f_{NRB} value is higher than the expected f_{NRB} values determined in Bailis Report[43] using the WISDOM method. While direct comparisons between the WISDOM and CDM methodologies are only sometimes appropriate —given the different approaches of the methodologies—the following factors can contribute to variations in the f_{NRB} estimates between the application of the CDM method in the present study and the WISDOM method in the Bailis Report[44]:

- **Parameter H (Total consumption of woody biomass in the applicable area in the relevant period (tonnes)):** The current report uses the most recent population statistics (2021) for Rwanda that represents an increase in population numbers since the 2009 data utilized in the Bailis Report. Hence, the value of parameter H is higher than the one used Bailis report as it is calculated using the total population.
- Updated 2019 FAO forest products statistics were used in the present study, whereas the comparison study used 2013 FAO forest products statistics.
- **Parameter RB (renewable biomass):** The approach used to determine the amount of forest that is accessible yields lower estimates in the present study than the comparison study. This results in a lower estimated RB and consequently increased NRB and f_{NRB}.

43 Bailis R, Drigo R, Ghilardi A and Masera O. 2015. The carbon footprint of traditional woodfuels. Nat Clim Chang 5: 266–272

44 Ibid.

- **Parameter MAI (Mean Annual Increment):** In the present study, MAIs were calculated using a weighted average based on the forest area of three categories (i.e., primary forests, above 20-year secondary forest, below 20- year secondary forest). Data from the 2019 Refinement of the IPCC Guidelines was used in combination with extrapolating the observed forest gain extents between 2000 and 2012 to a future 21-year period. As per the Bailis Report, MAI values were derived from a combination of field observations and IPCC values, followed by a different estimation of growth rates as a percentage of standing stock. This approach often yields higher MAIs and may lead to higher estimations of RB and subsequently, lower estimations of NRB and fNRB.

The calculated fNRB value of 0.91 indicates that the consumption of woody biomass from the accessible forest in Rwanda is greater than the country's capacity to renew the supply sustainably. This finding is supported by the consistent and widely-reported deforestation rate and dynamics observed for the country[45,46,47].

The f_{NRB} value of the project was also compared against other registered projects across VERRA and Gold Standard in Rwanda. The f_{NRB} value used by recently registered projects are as follows:

Registry	**Project ID**	**f_{NRB}**
Verra	2380	97%
Gold Standard	4261	98%
Gold Standard	2450	98%
Gold Standard	11205	90%
Gold Standard	2449	98%
Gold Standard	11098	96.82%
Gold Standard	12308	88.72%

The f_{NRB} value used in other registered projects across various standards range from 88.72% to 98%. As a measure of conservative, the fNRB value for the project has been changed to 88.72% which the lowest among the all the registered projects.

Step 6: Cross check in line with para 13 of CDM Tool 30 version 4.0

In line with para 13 of TOOL 30 V4, the NRB was compared to the top-down product of average above-ground biomass (146.06 t/ha) and the most recent annual deforestation data (3,187 ha/yr). NRB, as calculated in this report according to the CDM Tool 30 v4.0 2022, was found to be 5,598,738 t/yr, which is greater than the cross-check results based on deforestation (465,493 t/yr). It is expected that the cross-check estimate of biomass loss from deforestation

45 FAO. 2020b. Global Forest Resources Assessment 2020: Main Report. Food and Agriculture Organisation of the United Nations, Rome. Available at: https://www.fao.org/forest-resources-assessment/2020/en/

46 Global Forest Watch (GFW). 2023.

47 Mongabay. 2020.

could underestimate the extent of non-renewable biomass as it only considers the average biomass stocks from completely deforested areas. Unsustainable wood harvesting resulting in forest degradation and substantial reductions in biomass stocks, but not complete deforestation is not considered in the cross-check. Consequently, the estimate of NRB is considered applicable for this study given that the guidance provided by CDM Tool 30 v4.0 2022 was followed.